

HUMAN BIOLOGY ATAR
UNITS 3 & 4

UNIT 3

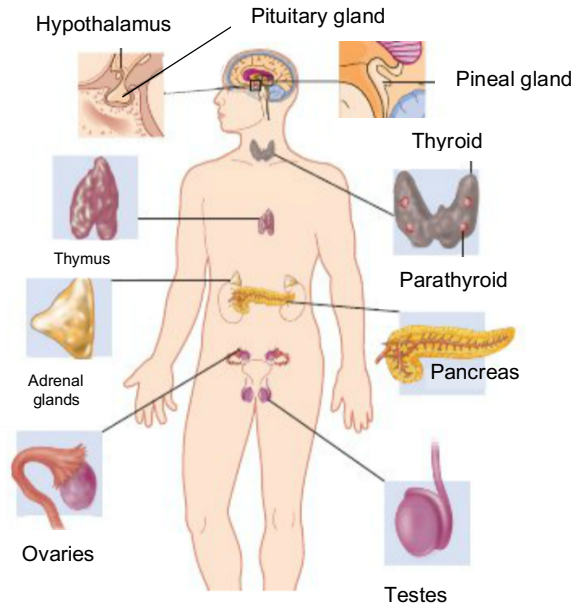
Endocrine System

Glands

- Endocrine and exocrine glands

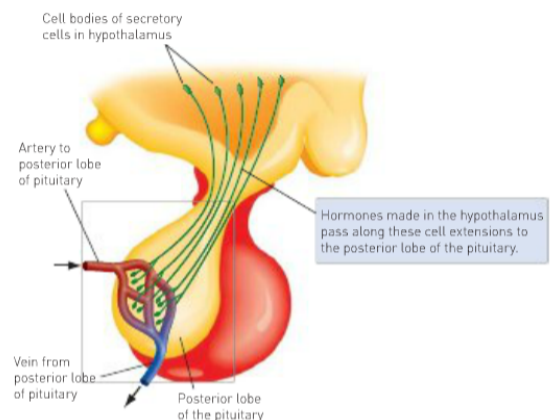
Exocrine glands: secrete into a duct that carries the secretion to the body surface or to a body cavity e.g. sweat glands, salivary glands

Endocrine (ductless) gland: secrete hormones into the surrounding extracellular fluid, which passes into the capillaries to be transported by the blood e.g. thyroid gland, pituitary gland



Endocrine glands

- Hypothalamus - located at the base of the brain, functions are carried out through the pituitary gland
- Pituitary lies just under the hypothalamus and is joined to it by the infundibulum
- Pituitary consists of anterior (front) and posterior (back) lobes – function separately
 - Anterior pituitary - has no nerves connecting it to the hypothalamus but is connected to it by a complex network of blood vessels.
 - Posterior pituitary – connected to hypothalamus by nerve fibres with cell bodies in hypothalamus. Release is stimulated by nerve contraction.



Hormone	Target Organ	Main Effects
Anterior Pituitary Gland		
Follicle-stimulating hormone (FSH)	Ovaries (females)	Growth of ovarian follicles
	Testes (males)	Production of sperm
Luteinising hormone (LH)	Ovaries (females)	Ovulation and formation of the corpus luteum
	Testes (males)	Stimulates interstitial cells of testes to secrete testosterone
Growth Hormone (GH)	All body cells	Growth and increase rate of protein synthesis

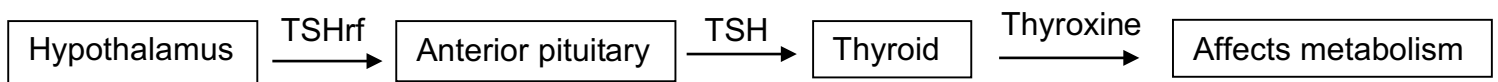
Thyroid-stimulating hormone (TSH)	Thyroid gland	Acts on the thyroid to stimulate release of hormones from thyroid
Adrenocorticotrophic hormone (ACTH)	Adrenal cortex	Acts on the adrenal cortex to stimulate the release of hormones from the adrenal cortex
Prolactin	Mammary glands	Milk production
Posterior Pituitary Gland		
Oxytocin	Uterus	Contraction of uterus during childbirth
	Mammary glands	Release of milk
Antidiuretic hormone (ADH)	Kidney tubules	Increase the permeability of the distal convoluted tubule to reabsorb water

Pineal Gland

- Found inside the brain
- Secretes melatonin
 - Regulates sleep patterns

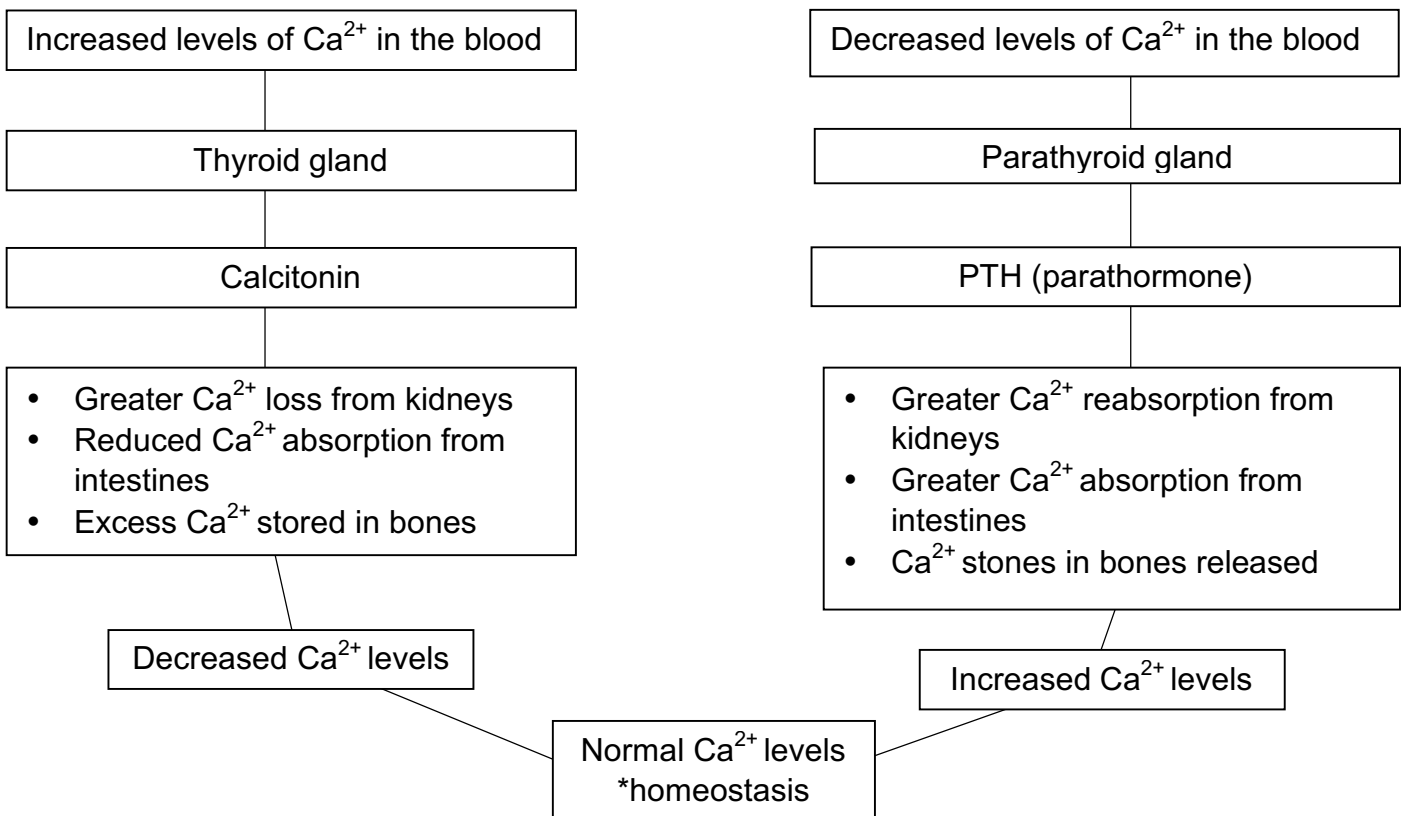
Thyroid

- Just below larynx, two lobes surrounding trachea
- Releases thyroxine
 - Controls body metabolism by regulating both catabolic and anabolic reactions
 - Brings about release of energy and maintains body temperature

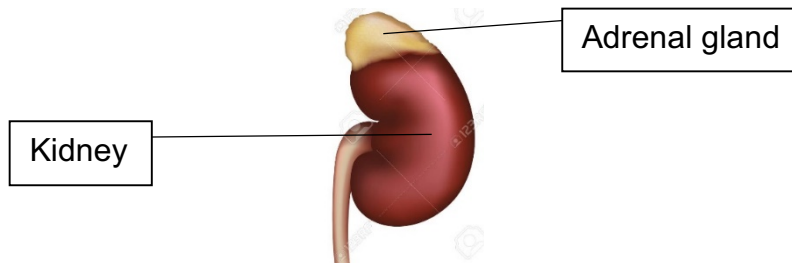


Parathyroid Glands

- Four pea-sized glands embedded at the back of the two lobes of the thyroid gland
- Parathormone (PTH) controls level of calcium in the blood



Adrenal Glands



- Two glands, one just above each kidney
- Each gland has an adrenal medulla (inner portion) and an adrenal cortex (outer portion)

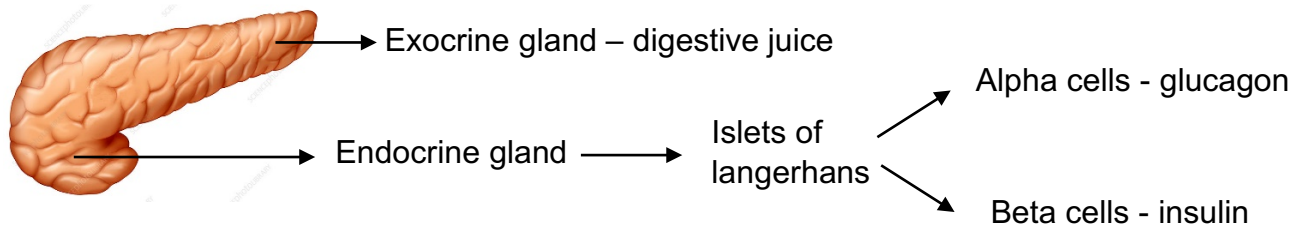
Adrenal Medulla

- Adrenaline (epinephrine) is similar to the effect of the sympathetic division of the autonomic nervous system (fight or flight response)
- Noradrenaline (norepinephrine) is similar to adrenaline but is particular to the heart rate and force of the heartbeat

Adrenal Cortex

- Produces >20 hormones called corticosteroids
- Aldosterone acts on the kidney to reduce the amount of sodium and increase amount of potassium in the urine
- Cortisol promotes normal metabolism, helps the body withstand stress and helps repair of damaged tissue

Pancreas



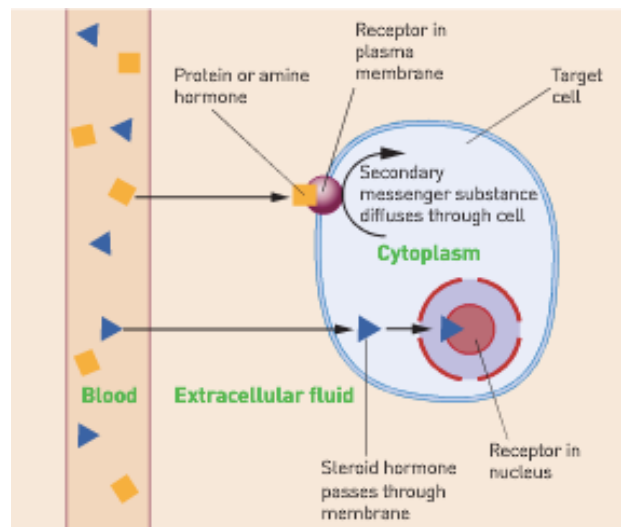
- Endocrine and exocrine gland
- Clusters of cells called islets of Langerhans
- Islets contain two types of cells
 - Alpha cells
Glucagon: increases blood glucose levels
Promoting the breakdown of glycogen to glucose in the liver (glycogenolysis)
Breaking down fat in the liver and adipose tissue to convert it to glucose (gluconeogenesis)
 - Beta Cells
Insulin: decreases blood glucose levels
Promoting uptake of glucose from the blood by body cells
Liver converting glucose to glycogen (glycogenesis)
Formation of glycogen from glucose in skeletal muscle.

Hormone Summary

Gland	Hormone	Production site	Target Cells	Effect
Anterior Pituitary Gland	Follicle-Stimulating Hormone (FSH)	APG	Ovaries, testes	Growth of ovarian follicle, production of sperm
	Luteinising Hormone (LH)	APG	Ovaries, testes	Ovulation and corpus luteum, secretion of testosterone
	Growth Hormone (GH)	APG	All body cells	Growth, increase rate of protein synthesis
	Thyroid-Stimulating Hormone (TSH)	APG	Thyroid gland	Secretion of hormones from thyroid
	Adrenocorticotrophic Hormone (ACTH)	APG	Adrenal cortex	Secretion of hormones from adrenal cortex
	Prolactin	APG	Mammary glands	Milk production
Posterior Pituitary Gland	Oxytocin	Hypothalamus	Uterus and mammary glands	Contraction of uterus during childbirth, release of milk
	Anti-diuretic Hormone (ADH)	Hypothalamus	Kidney tubules	Increase permeability of DCT and collecting duct to reabsorb water
Thyroid	Thyroxine (T3 and T4)	Thyroid	Most cells	Increase metabolism and therefore oxygen consumption and heat production
	Calcitonin	Parathyroid	Bones, kidney, blood, intestines	Decrease Calcium levels
Para-thyroids	Parathormone (PTH)	Parathyroid	Bones, kidney, blood, intestines	Increase Calcium levels
Adrenal Glands	Aldosterone	Adrenal Cortex	Kidney	Increase reabsorption of sodium ions and decrease reabsorption of potassium ions
	Cortisol	Adrenal Cortex	Most cells	Promotes normal metabolism, withstand stress, replace damaged tissue
	Adrenaline (epinephrine)	Adrenal medulla	Most cells	Reinforces the effect of sympathetic nervous system (fight or flight)
	Noradrenaline (norepinephrine)	Adrenal medulla	Most cells	“ “
Pancreas	Glucagon	Alpha cells	Most cells	Decrease glucose absorption therefore increasing BGL
	Insulin	Beta cells	Most cells	Increase glucose absorption therefore decreasing BGL

Types of hormones

- Hormones do not start reactions, but they regulate reactions
- Hormones are required in minute quantities
- Hormones from animals can be transferred
- Hormones cannot be stored, they have an expiry date – in our body the liver catabolises them
- May be proteins, steroids or amines
- Proteins and amines attach to a receptor molecule on the membrane of the target cell
 - Causes a secondary messenger substance to diffuse through the cell and activate particular enzymes
- Steroid hormones work by entering target cells and combining with a receptor inside the cell
 - Activates the genes controlling the production of particular proteins



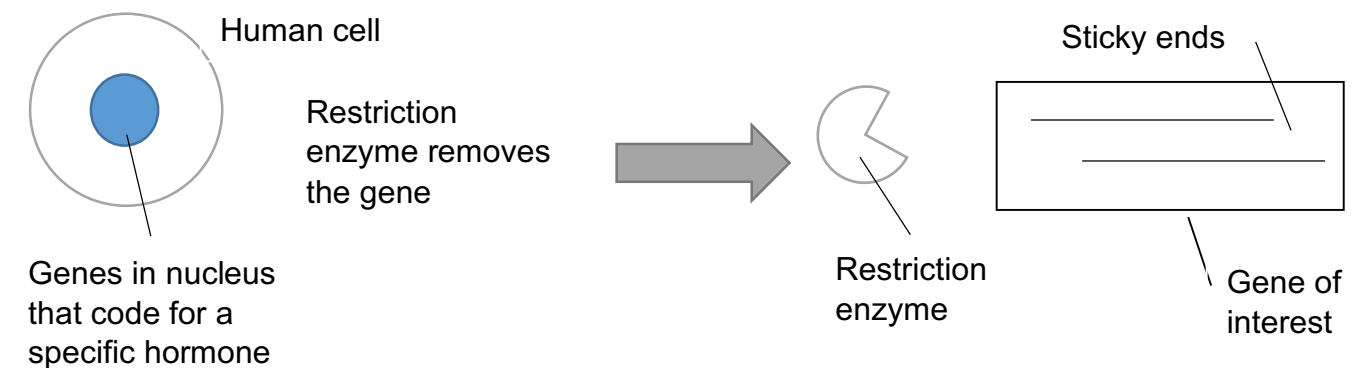
Enzyme Amplification

- One hormone molecule can manufacture or activate thousands of enzyme molecules
- Hormone triggers a cascading effect called enzyme amplification
- A very small stimulus can cause a very large effect

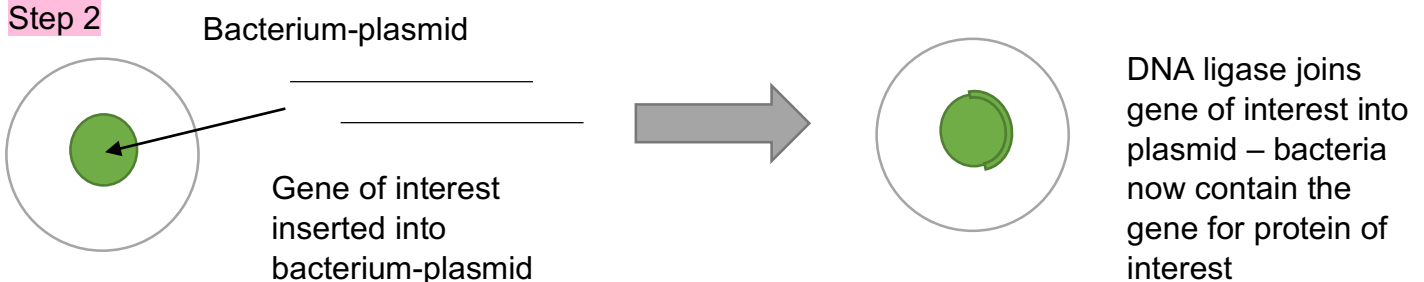
Synthesising Human Proteins

- Done using recombinant DNA technology
- Synthesis of human insulin using recombinant DNA technology:

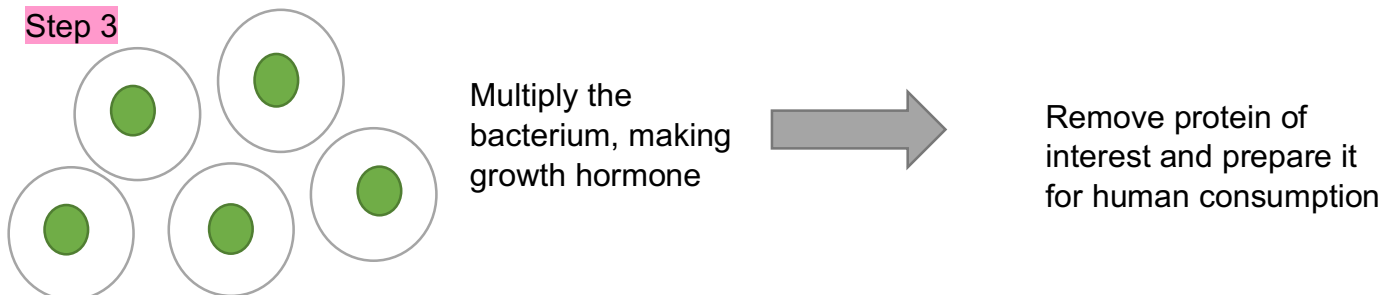
Step 1



Step 2



Step 3



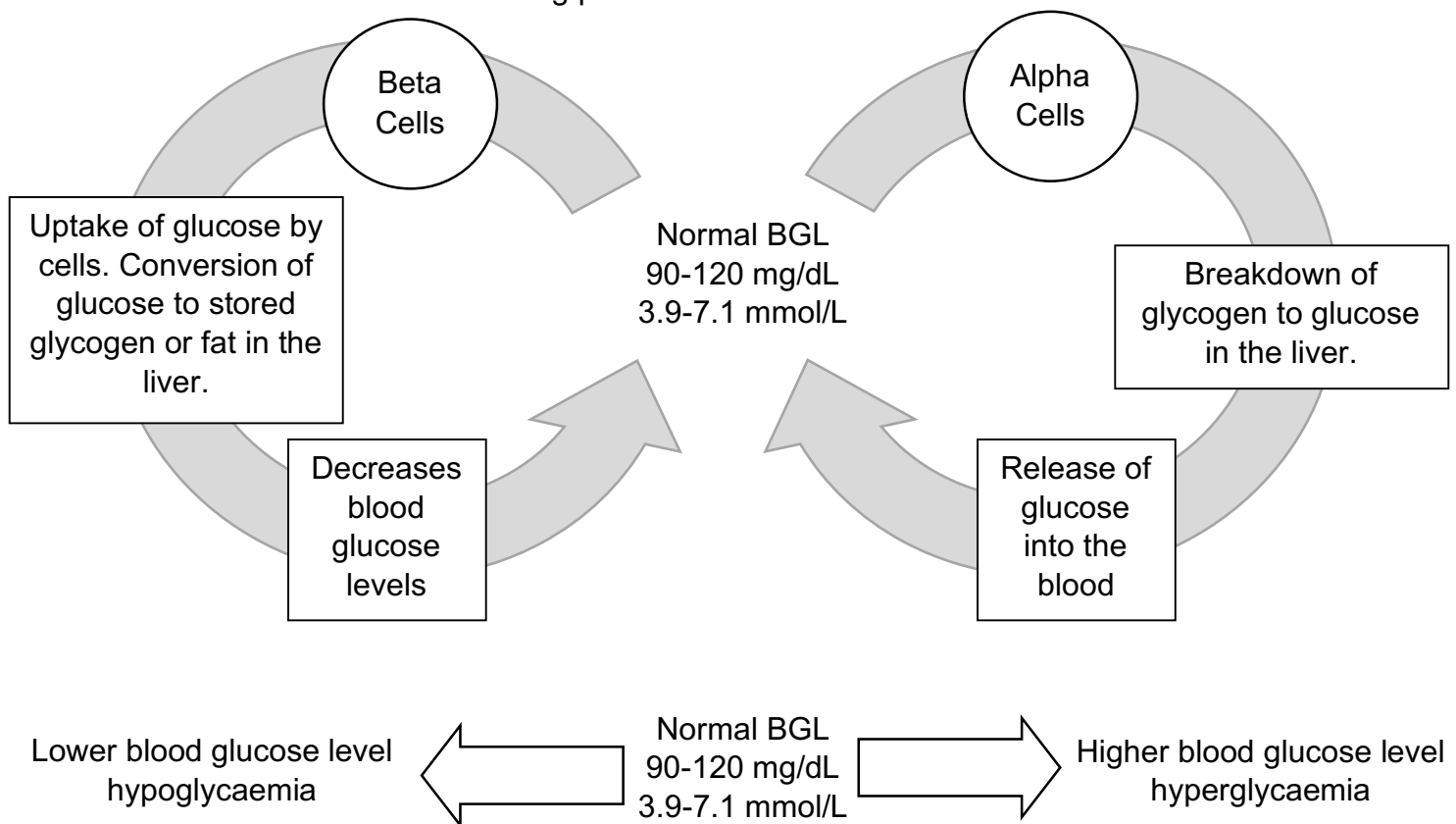
	Type 1 Diabetes	Type 2 Diabetes
Cause	• Deficiency of insulin • Lack of insulin production (beta cells destroyed in autoimmune reaction)	• Insulin receptors are resistant to the hormone – cell becomes impermeable to glucose • Cells can't uptake glucose
Symptoms	• Severe • Hyperglycaemia • Glucose in urine • Increased urine production • Excessive thirst and hunger • Weight loss • Ketosis	• Increased appetite (overeating may contribute to existing obesity problem) • Urine production increases • Glucose in urine • Muscle weakness, fatigue • Frequent infections, poor wound healing
Treatment	Present: • Regular insulin injections • BGL monitoring New: • Cell therapy (transplant of beta cells)	• Regularly check blood glucose level • Manage diet to reduce fluctuations in blood glucose level • Reduce weight (exercise, diet) • Reduce blood pressure

	Future: Gene therapy (gene for insulin inserted into diabetic's cell)	- Take anti-diabetic drugs - Insulin therapy
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	Hyperthyroidism	Hypothyroidism
Cause	Too much thyroxine	Too little thyroxine
Symptoms	- Weight loss - Appetite change - Muscle weakness - Grave's disease (the eye thing)	- Slow metabolism - Weight gain - Swelling - Enlarged thyroid gland (goiter)
Treatment	Radioactive iodine	- Synthetic thyroxine (levothyroxine) - Increased iodine intake

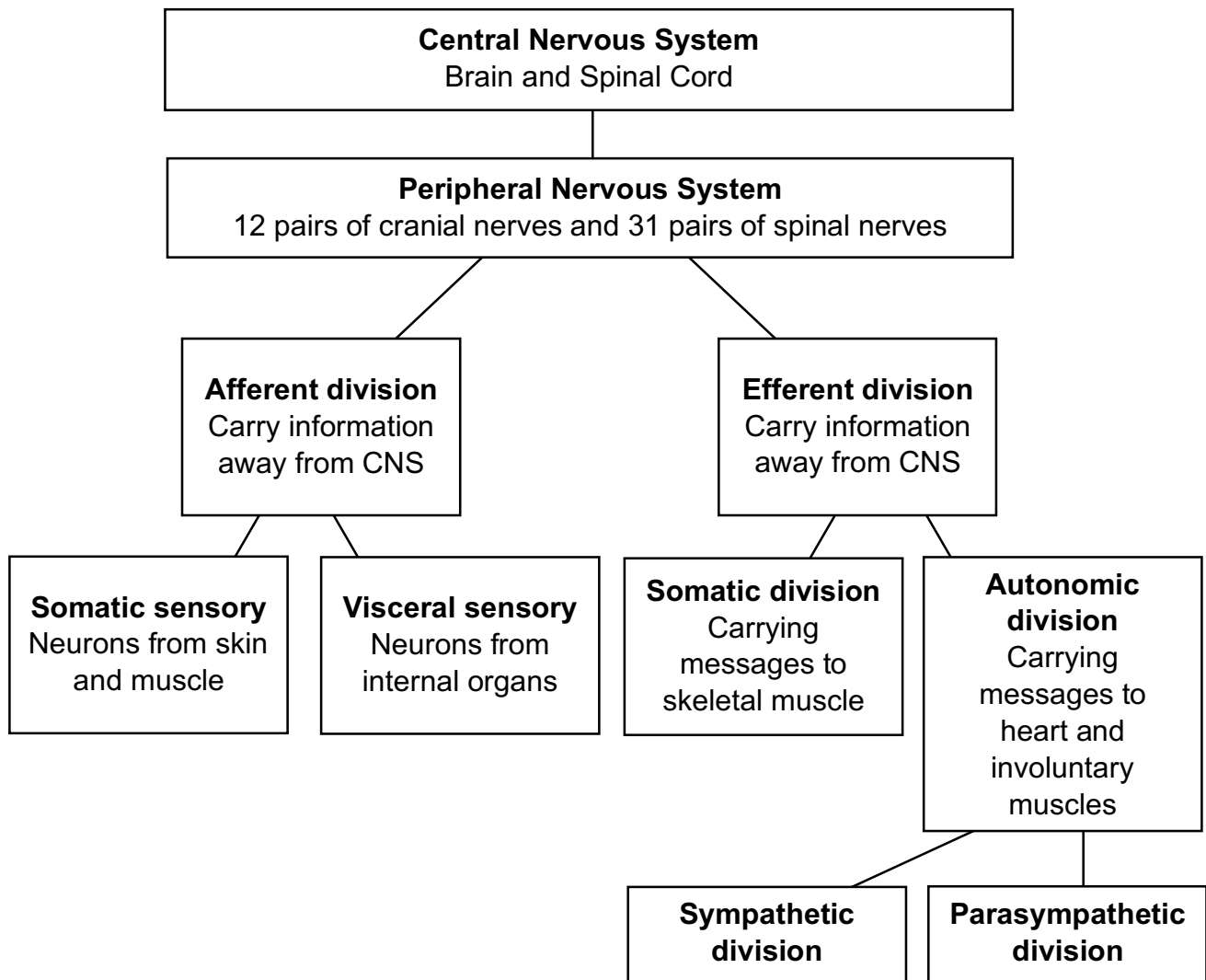
Control of Blood Glucose Levels

- Glucagon and Insulin mediate the regulation of blood glucose levels
- Negative feedback loops
- Glucagon – increase blood glucose levels, synthesis of glucose from amino acids
- Insulin – decrease blood glucose levels, synthesis of glycogen and cellular uptake of glucose
- When food is unavailable for long periods of time



Nervous System

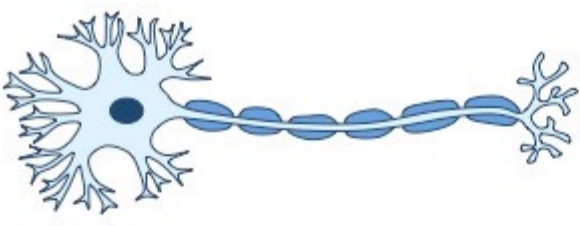
- Communication network and control centre of the body
- Maintains homeostasis
- Two main parts
 - Central Nervous System (CNS) consists of brain and spinal cord
 - Peripheral Nervous System (PNS) is the nerves that connect the CNS with the receptors, muscles and glands



Neurons

Term	Definition
Neuron	Single nerve cell
Nerve-fibre	Extension of the cell, usually an axon
Synapse	Junction between the branches of two neurons. Is a gap, the signal is carried across the gap by a neurotransmitter
Neuromuscular junction	Axon terminal joins a muscular fibre

Types of Neurons

Type of Neuron	Diagram	Purpose
Multipolar Motor neuron, efferent neuron		• Takes impulses away from the CNS • EFFerent for f off
Bipolar Sensory neuron, afferent neuron		• Takes impulses into the CNS
Unipolar Sensory neuron, afferent neuron		• Takes impulses into the CNS

Transmission of Impulses

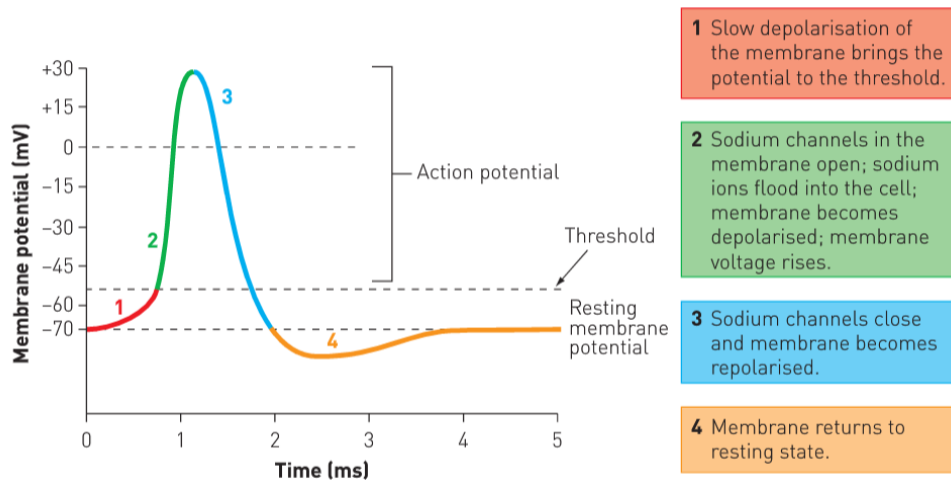
Transmission of Nerve Impulses Across an Unmyelinated Neuron

Resting Potential

- Normal state, not stimulated
- Inside of membrane is negative compared to outside which is positive
- Difference is called resting potential
- Resting potential is -70mV
- Difference in concentration of ions (K^+ , Na^+ , Cl^-) maintained by the active transport of ions using carrier-mediated transport (pumps)
- Sodium-potassium ion pump
- These pumps move $3Na^+$ out of the cell for every $2K^+$ ions into the cell
- Requires ATP

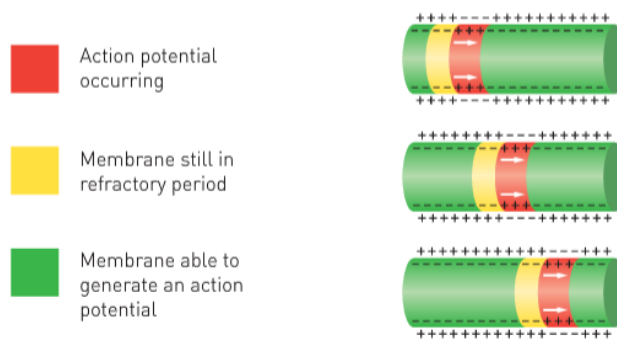
Action Potential

- When the membrane receives stimulation, the negative charge inside the membrane becomes positively charged
- As a result of action potential, the membrane is depolarised
- In order for the neuron to be stimulated, it has to be above the threshold -55mV
- Caused by an influx of Na^+
- ALL or NOTHING response – it either happens or it doesn't
- Na^+ gates open rapidly, influx of Na^+ by diffusion
- Depolarises the membrane
- K^+ gates take longer to open but eventually do
- At about 30mV the Na^+ entering are equal to K^+ leaving, no net movement with Na^+ gates shut and the K^+ gates open the membrane begins repolarising
- Does overshoot called hyperpolarisation



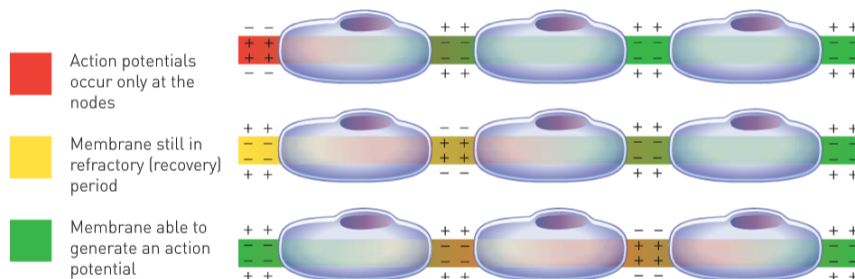
Refractory Period

- After an action potential, K^+ ions quickly restore to resting potential
- Na^+ diffusion into the cell is prevented in that region
- During the 1ms afterward no action potentials can be generated
- Therefore an action potential can only travel in one direction



Transmission of Nerve Impulses Across a Myelinated Neuron

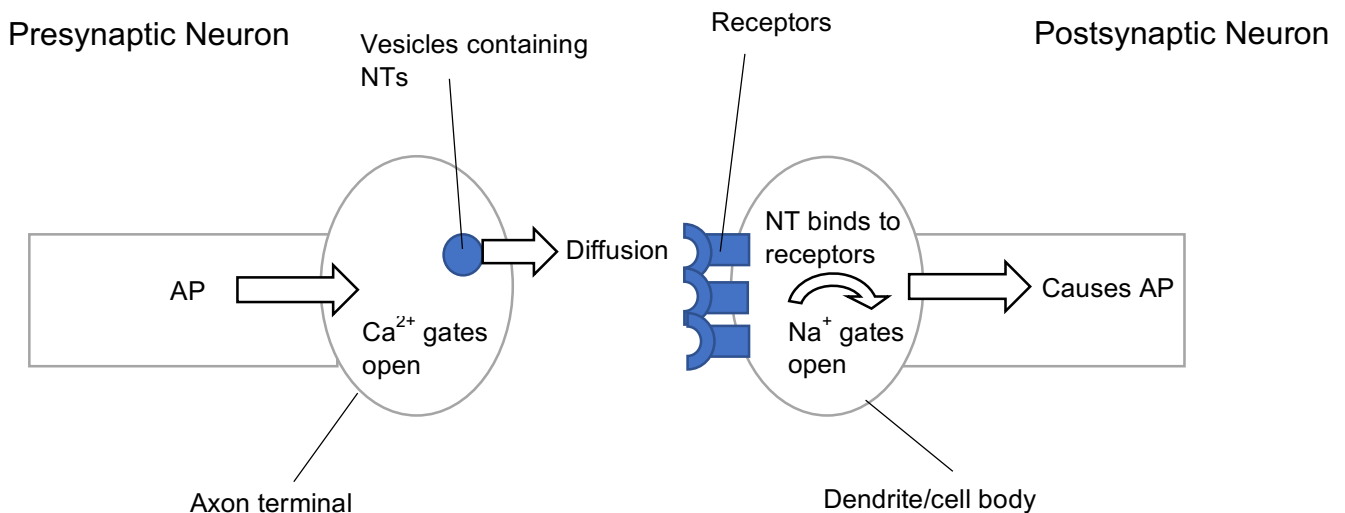
- Called saltatory conduction
- Action potential moves from one node of ranvier to the next
- Moves 140m/s



Transmission across a synapse

- In a synapse there is no membrane
- Neurotransmitters diffuse across gap and attach to receptors
- Over 100 neurotransmitters (NTs)
 - Acetylcholine
 - Adrenaline and Noradrenaline
 - Dopamine
- Only occurs from axon terminals to dendrite/cell body

Transmission across the synapse



Excitatory	Inhibitory
• Increase stimulation of transmission at synapse ○ Acetylcholine (Ach) ○ Adrenaline ○ Noradrenaline • Nerve agents (organophosphate) stops enzyme (denatures the enzyme) from breaking down Ach • Causes constant muscle contraction i.e. seizures, difficulty breathing, loss of control, urination and defaecation, spasms and pain, death	• Increase depression of transmission at synapse ○ Snake venom ○ Spider venom • Snake venom inhibits Ach from binding to receptors, muscle contraction does not occur • Causes paralysis

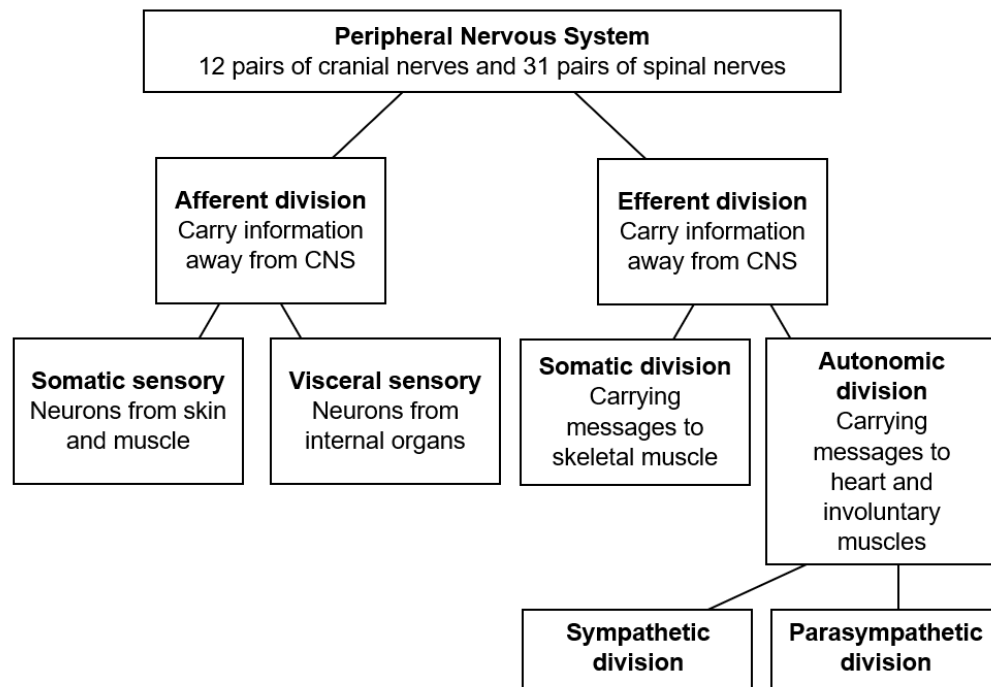
Divisions of the Nervous System

Peripheral Nervous System (PNS)

- Consists of nerve fibres which carry information to and from the CNS
- Consists of groups of nerve cell bodies called ganglia
- Nerve fibres are arranged into nerves which arise from brain and spinal cord
- 12 pairs of cranial nerves (brain)
 - Optic nerve, auditory nerve
 - Usually missed nerves being able to carry information to the brain (sensory fibres) and those that carry impulses away from the brain (motor fibres)
 - A few cranial nerves carry only sensory or motor impulses
- 31 pairs of spinal nerves (spinal cord)

Spinal nerves

- 8 pairs of cervical spinal nerves
- 12 pairs thoracic spinal nerves
- 5 pairs of lumbar spinal nerves
- 1 pair of coccyx spinal nerve



Afferent division of the PNS

- Carries impulses from sensory receptors to the CNS (away from CNS)
- Afferent nerve cell can be: somatic sensory and visceral sensory
 - Somatic: receptors in skin, around muscles and joints
 - Visceral: receptors in internal organs

Efferent division of the PNS

- Carries impulses away from the CNS
- Somatic division takes impulses from the CNS to skeletal muscles
- Autonomic division divided into sympathetic and parasympathetic divisions

Autonomic Division

- ANS responsible for our internal environment by controlling homeostatic mechanisms without conscious control
 - Heart rate, blood pressure, temperature, digestion, pupil size, airflow, defaecation and urination
- Carries messages away from the CNS
- Two types of neurons
 - First cell body in grey matter of CNS
 - Second has cell body in ganglion
- Is divided into Sympathetic and Parasympathetic divisions
- Parasympathetic: Tends to control homeostatic mechanisms
 - **Rest and digest**
- Sympathetic: Takes control during strenuous physical activity
 - **Fight or flight** – stimulated by fear, anger, competition or stress
 - Rate and force of heart beat and blood pressure, dilate blood vessels, non-essential organ blood flow restricted, dilated airways and increase depth of breathing, blood glucose level increase, sweat secretion increase, Adrenal Medullae release adrenaline and noradrenaline

Central Nervous System

- Consists of the brain and spinal cord
- Control centre of the nervous system

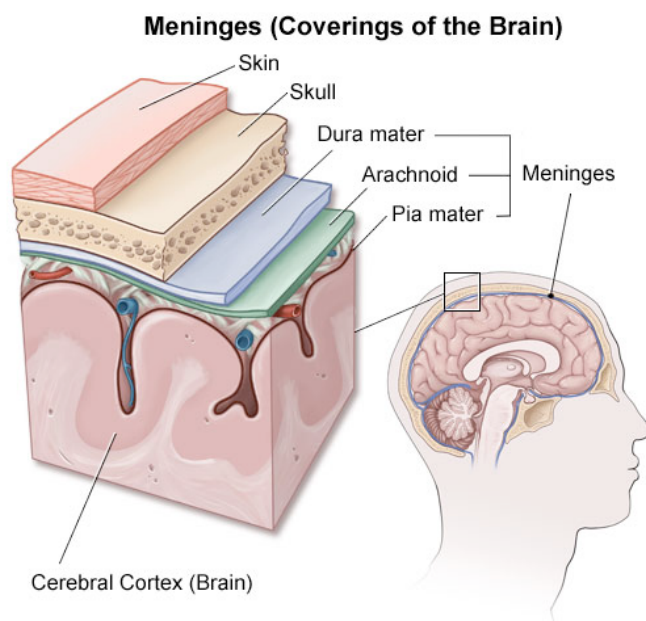
Protection of the CNS

1. Bone

- Cranium protects the brain
- Spinal cord protected by vertebrae
 - Runs through vertebral canal

2. Meninges

- Three layers of connective tissue forming membranes that cover the surface of the brain and spinal cord



Pia Mater

Inner layer

Delicate, has blood vessels

Arachnoid

Middle layer

Loose mesh of fibres

Dura Mater

Outer layer

Tough and fibrous

3. Cerebrospinal Fluid

- Clear, watery fluid that occupies space between middle and inner meninges, cavities in the brain and centre of spinal cord
- Contains water, cells, glucose, urea and salts

PST

Protection: shock absorber to cushion blows/shocks

Support: Brain suspended in cranium and floats in fluid

Transport: Takes nutrients to and wastes away from the cells of the brain and spinal cord

Brain

Diencephalon

Thalamus

Hypothalamus

- HR, BP, digestive juices, alimentary canal, diameter of pupils
- Body temp., food+water intake
- Hormones and endocrine system

Pituitary gland

Corpus Callosum

- Wide band of nerve fibres
- Lies underneath the cerebrum
- Nerves cross from 1 hemisphere to the other
- Allows communication between hemispheres

Cerebrum

- Highly folded = large surface area
- Fissure: longitudinal fissure that separates brain into two hemispheres
- Fissures and sulci divide brain into four hemispheres
- Frontal, parietal, occipital and temporal lobes
- Voluntary movement

Midbrain

Pons

Ventricle and cerebrospinal fluid

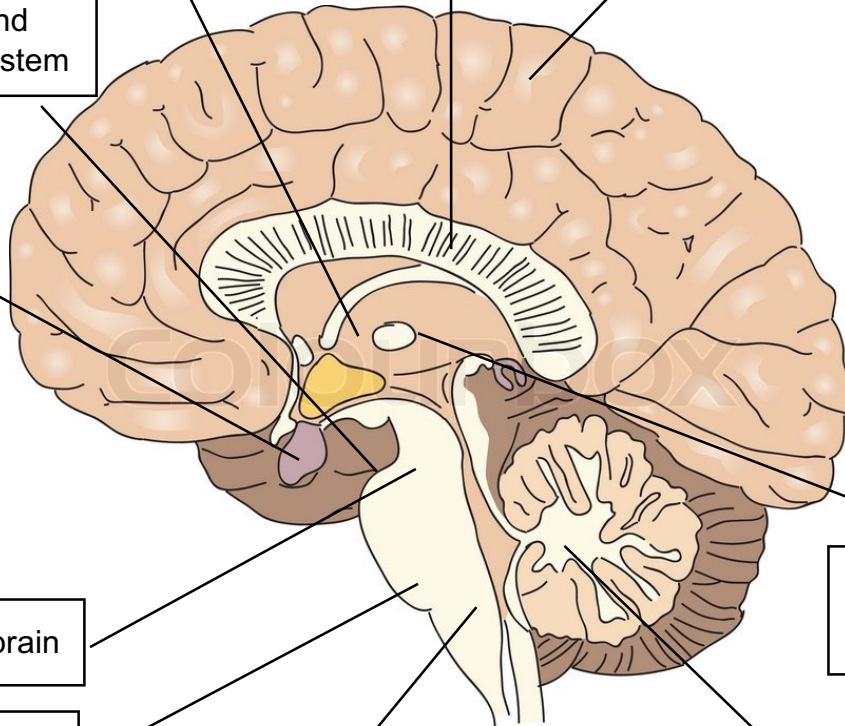
Brain stem

Medulla Oblongata

- Continuation of spinal cord
- Automatically adjusts body functions as it contains the cardiac, respiratory and vasomotor centres
- All functions under the influence of the hypothalamus

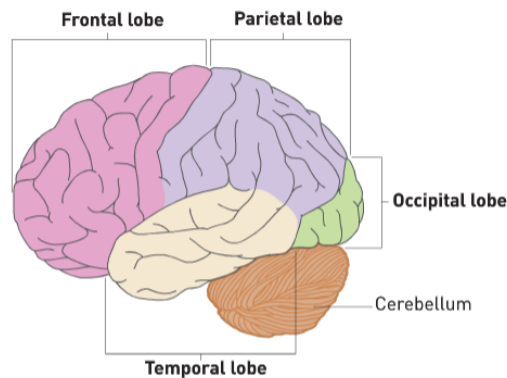
Cerebellum

- Second largest part of brain
- Surface folded into a series of parallel ridges
- Control of posture and balance; information received from inner ear and stretch receptors in skeletal muscles
- Fine coordination of muscle movement
- White matter inside, grey matter outside



Cerebrum

- Rounded ridges – convolutions (gyri) separated by either shallow downfolds (sulci) or deep down folds (fissures)
- Four lobes



- 3 different areas: sensory, motor, association

Sensory areas

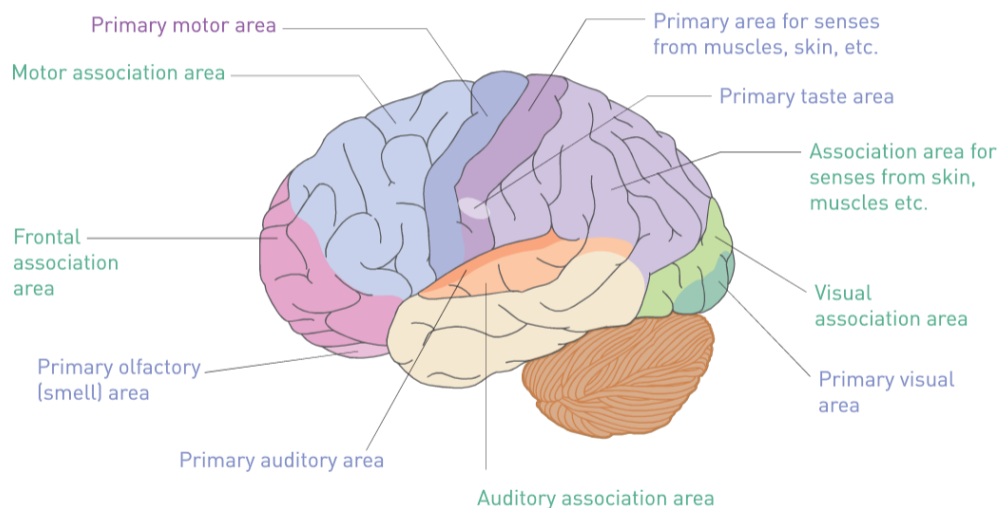
Receive and process nerve impulses from the senses

Motor areas

Send impulses to muscles, especially voluntary

Association areas

Interpret information from sense and make it useful



Control of Movement

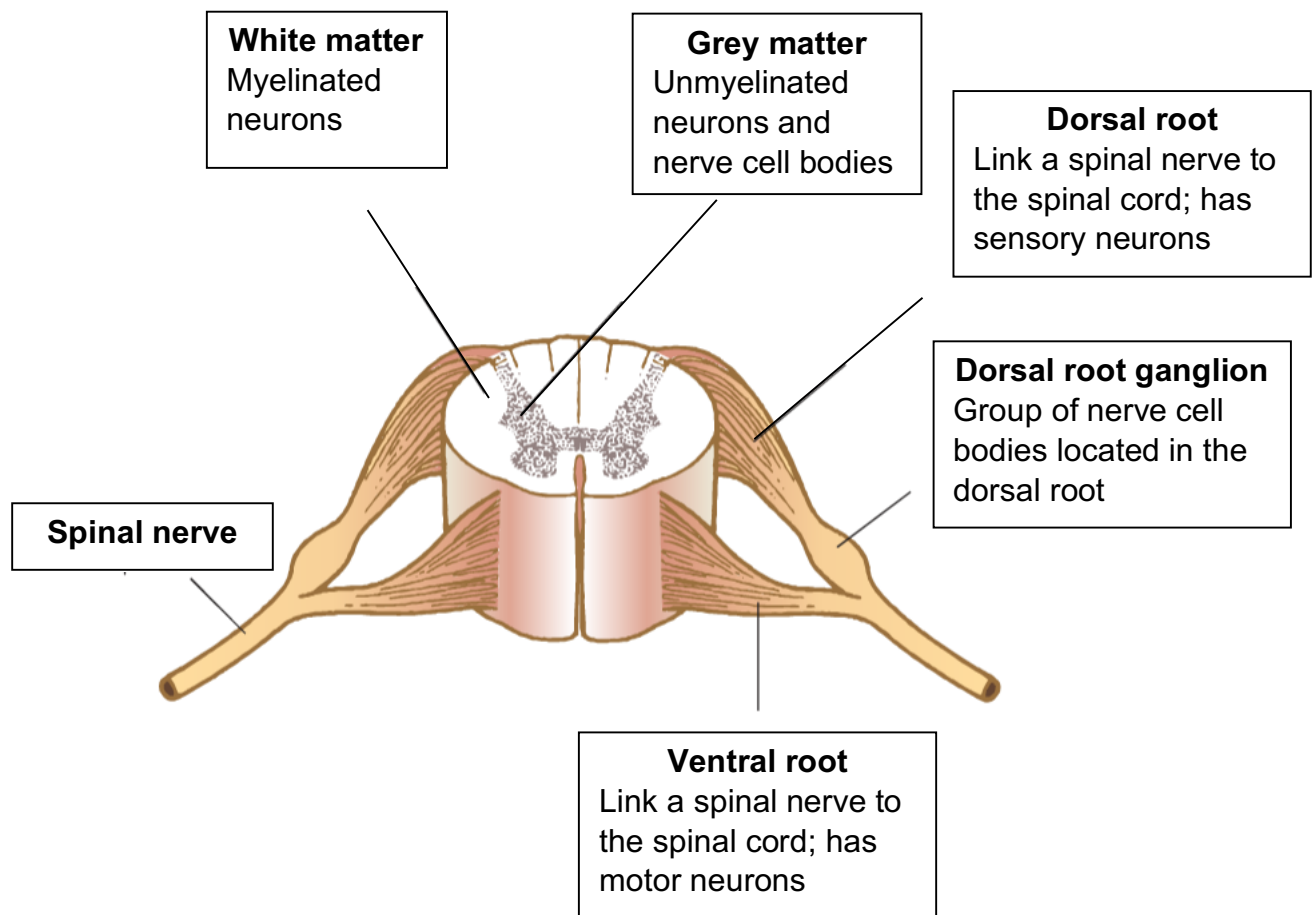
Cerebrum	Cerebellum
• Intention to contract muscles begins in the motor association area • Plans the behaviour and makes up the sequence and intensity of the muscle contractions required • Program sent to the primary motor area • Nerve impulses sent to muscles • Upper motor neurons carry impulses to appropriate level of spinal cord (neurons cross over) • Lower motor neuron carry impulses to the muscles	• Achieves smooth and coordinated movement and maintains pressure and muscle tone during movement by integrating movement from: ○ Cerebrum: position of body ○ Semi-circular canals (inner ear): movements of the head ○ Saccule and utricle (inner ear): position of the head ○ Eyes: observe body's position in space ○ Pressure receptors in skin: relative amount of pressure on parts of body ○ Stretch receptors in muscles and joints: body movement and joint position

Explain how the cerebrum and cerebellum initiate and control movement of skeletal muscle. (4 marks)

- Cerebrum initiates movement through the motor association area which initiates the primary motor area. The primary motor area sends an action potential via upper motor neurons to the spinal cord. This action potential affects the lower motor neuron and the action potential reaches the neuromuscular junction and muscle contraction occurs
- Cerebellum controls fine coordination of muscle movement through information received from sensory receptors of the ear and stretch receptors in skeletal muscle

Reflex Arcs

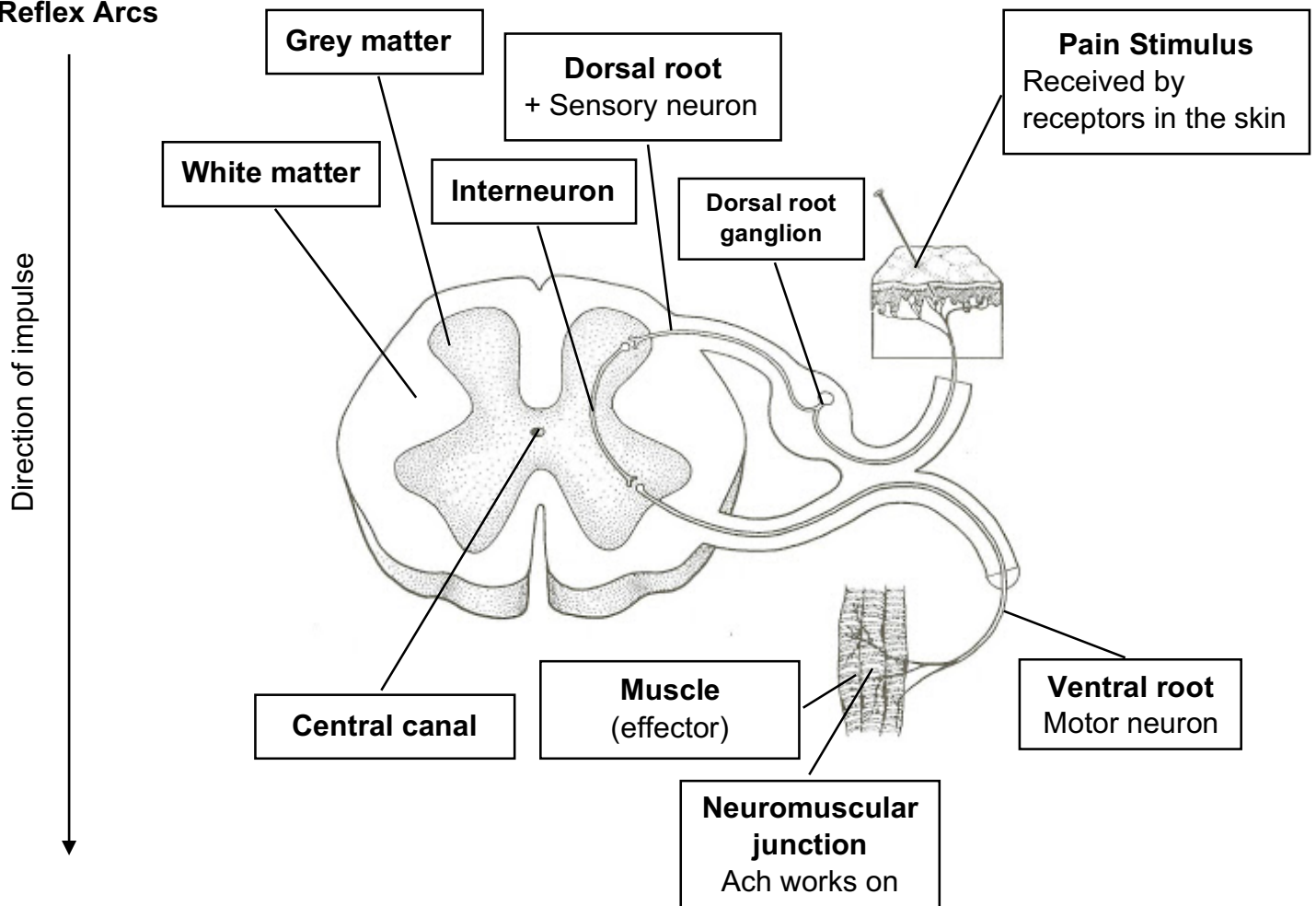
Spinal Cord



Functional properties of reflex arcs

- Involuntary – no thought needed for the reflex to occur
- Stereotypical – the arc happens the same way every time
- Rapid – the stimulus is changed as soon as possible

Reflex Arcs



Autonomic division

Structure	Sympathetic Fight or flight	Parasympathetic Rest and digest
Heart	Increase cardiac output $CO = \text{Stroke volume} \times \text{heart rate}$	Decrease cardiac output
Eyes	Dilates pupils	Constricts pupils
Lungs	Dilates bronchioles Increase breathing rate	Constricts bronchioles Decrease breathing rate
Bladder and anus	Contracts sphincters	Relaxes sphincters
Saliva production	Decreases saliva	Increases saliva
Liver and gall bladder	Increase breakdown of glycogen → glucose (glycogenolysis) Inhibits gall bladder	Increase conversion of glucose → glycogen (glycogenesis) Stimulates gall bladder
Stomach and intestines	Inhibited movement	Stimulates movement
Adrenal Medulla	Stimulates hormone secretion Adrenaline and Noradrenaline	No obvious effect
Sweat glands	Increase sweat production	No obvious effect
Skin	Vasoconstriction	Little effect
Skeletal muscles	Vasodilation of blood vessels	No effect
Internal organs	Vasoconstriction of all except heart and lungs	Little effect

Receptors

Type of Receptor	Stimulus	Location
Thermoreceptor	Temperature	Skin, hypothalamus
Nocireceptors/ mechanoreceptors	Intense stretch or strain, damage or threat of damage	Skin
Baroreceptors	Pressure in blood vessels	Aortic and carotid bodies
Osmoreceptors	Osmolarity: concentration of substances dissolved in the water of blood plasma	Hypothalamus
Proprioceptors	Sense of position	Inner ear, muscles, tendons, joints
Stretch receptors	Relative position	Muscles
Chemoreceptors	Presence of chemicals	Aortic and carotid bodies
Pain receptors	Pain, damage or threat of damage	Skin
Hair root plexus	Movement of hair	Hair follicles (skin)

Nervous vs. Endocrine

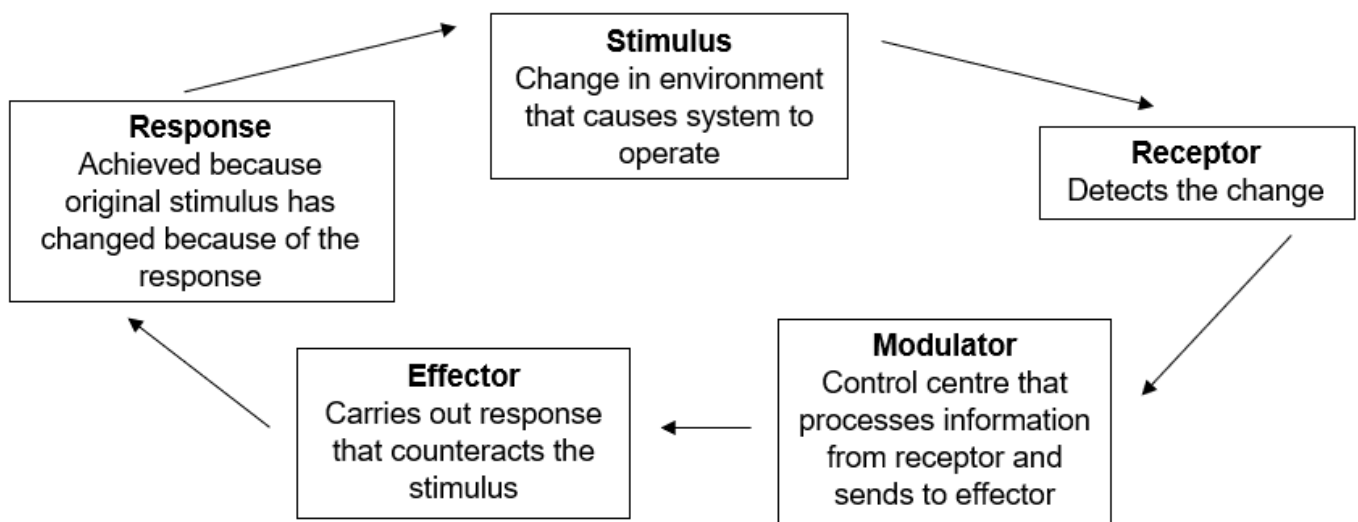
Structure	Nervous System	Endocrine System
Speed of Action	Rapid – within milliseconds	Slower – can take seconds to days for action
Duration of Action	Brief – usually short lived	Longer lasting – usually continues after stimulus has stopped
Nature	Electrochemical – Electrical impulse (sodium-potassium pump) and neurotransmitters	Chemical messengers - hormones
Transmission	Along the membrane of neurons	Blood stream
Specificity	Localised specificity – very specific	Wide spread specificity – can affect many cells, general

Homeostasis

Homeostasis: the maintenance of a constant internal environment where cells can carry out their specific functions as efficiently as possible.

Feedback Systems

- Body responds to stimulus → response alters original body stimulus
- Two types of feedback loops
 - Positive: The response reinforces the stimulus
e.g. Oxytocin
 - Negative: The response counteracts the stimulus
e.g. Anti-diuretic hormone



Thermoregulation

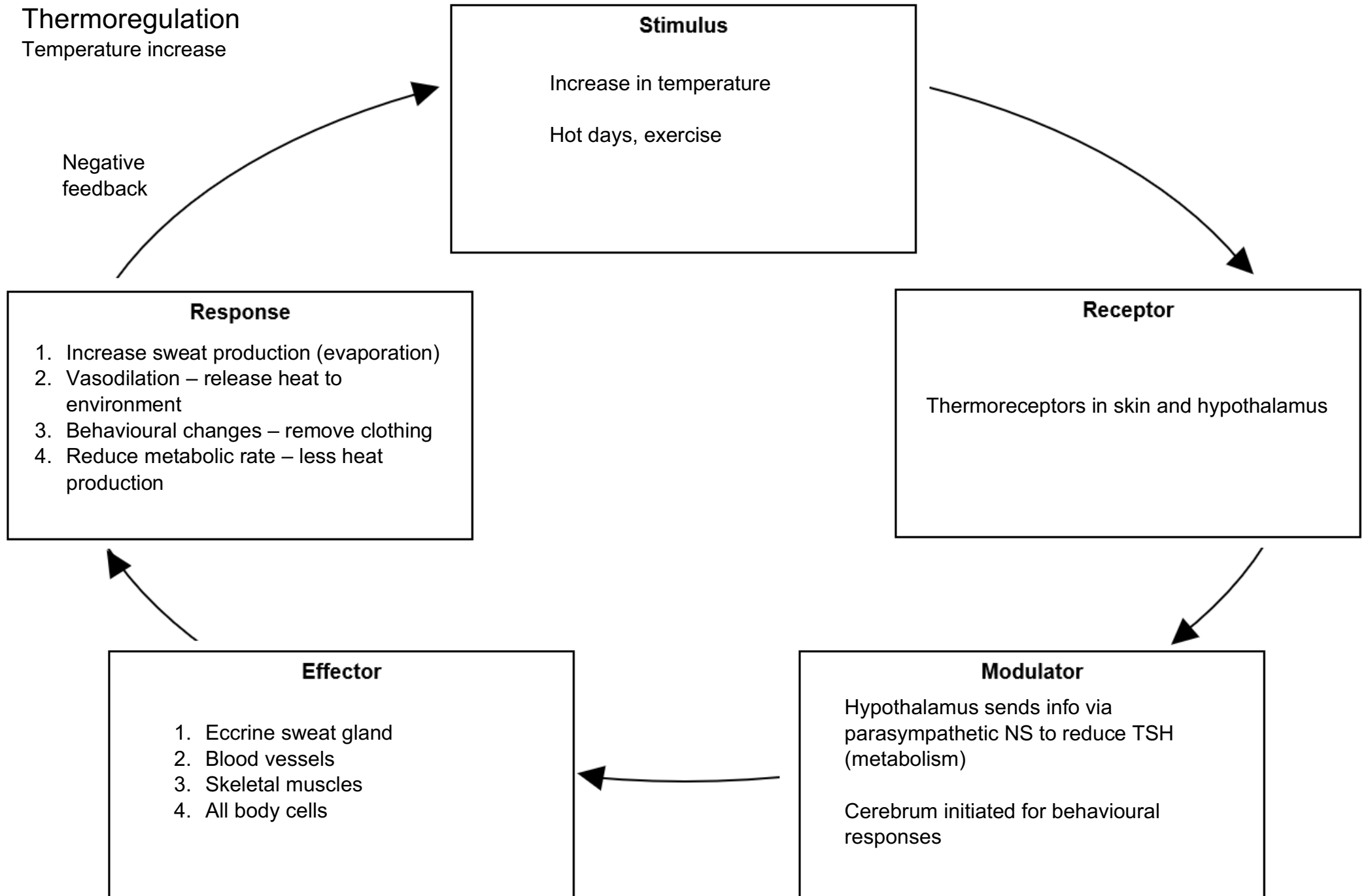
Rise in body temperature		Fall in body temperature	
Increase heat loss	Decrease heat production	Decrease heat loss	Increase heat production
Vasodilation Body increases blood flow to the skin, allows heat loss via radiation, prompts cooling of the blood that is flowing through the skin Sweating Sweat secreted from sweat glands to skin, contains salt, urea, lactic acid, K^+ ions and water. Cools by evaporation, environmental temperature has to be greater than body temperature to be effective. Behavioural response Decreasing activity, air conditioning, removing clothing Surface area Increase surface area by spreading out	Activity Decrease in voluntary activity, meaning there is a decrease in cellular respiration (heat is a product) Metabolic rate A reduction in the secretion of thyroxine decreases the metabolic rate, meaning there is a decrease in cellular respiration (heat is a product)	Vasoconstriction Body decreases blood flow to the skin, which decreases the transfer of heat from internal body organs to the skin, allowing less heat to be lost from the body surface Sweating Reduction in sweating Behavioural response Putting on more clothes, using a heating device Surface area Reduction of surface area by curling into a ball or huddling Pyloerection Goosebumps – hairs stand on end to keep heat in	Shivering Hypothalamus stimulates part of the brain that increases skeletal muscle tone. This increases body heat production Activity Increase in voluntary activity, meaning there is an increase in cellular respiration (heat is a product) Metabolic rate An increase in the secretion of thyroxine increases the metabolic rate, meaning there is an increase in cellular respiration (heat is a product)

Heat Transfer

Conduction Contact between the skin and another surface	Convection Heat through the air, e.g. wind
Evaporation We all know what evap is but can't define it	Radiation Like from a fire

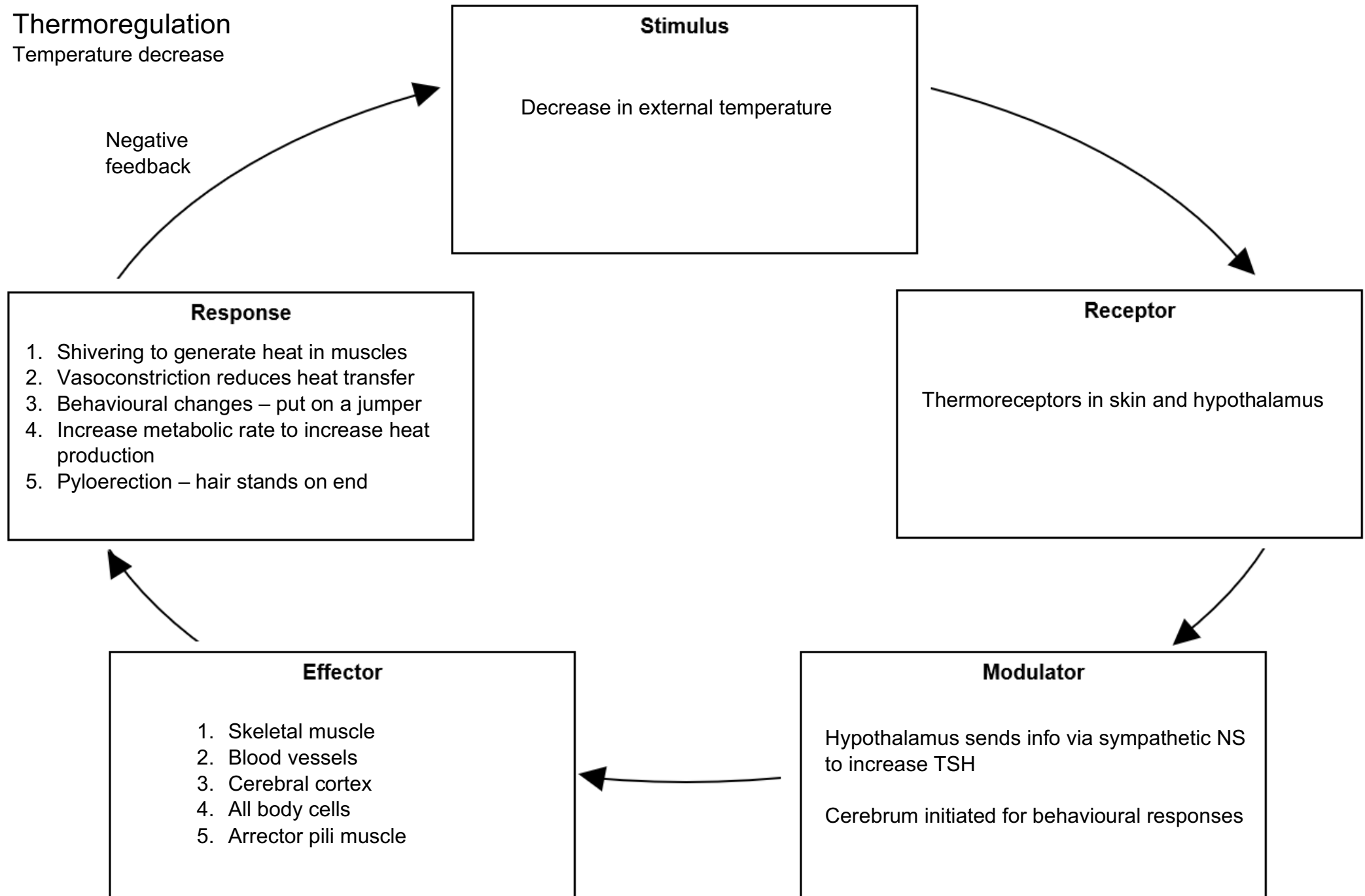
Thermoregulation

Temperature increase



Thermoregulation

Temperature decrease



Blood Sugar Levels

- Glucose energy source of cell – correct amounts must be maintained to supply these cells
- BGL is maintained by storing unused glucose as glycogen
- BGL (blood glucose level) is coordinated by the liver, pancreas and adrenal glands
- Liver is able to store 100g of glycogen
- Skeletal muscle is able to store 500g of glycogen
- Blood travels to the liver from the intestines
 - Glucose used by the liver
 - Stored as glycogen
 - Continues to circulate
 - Converted to fat for long term storage

Glycogenesis: Synthesis of glycogen from glucose

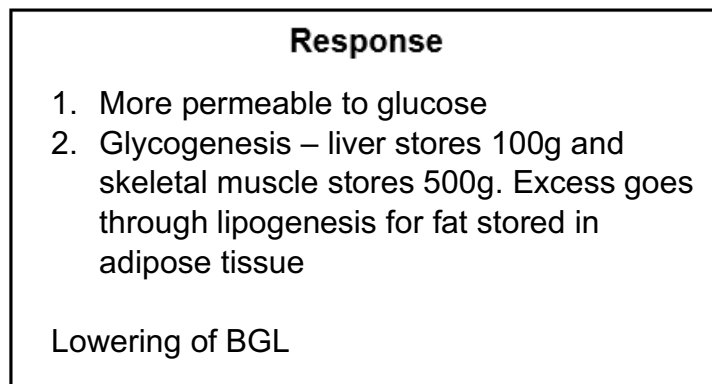
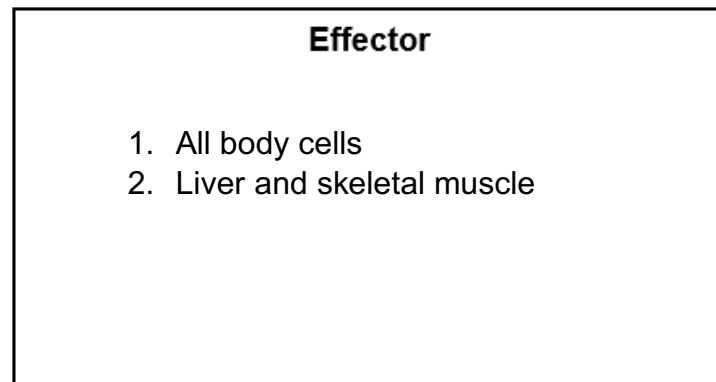
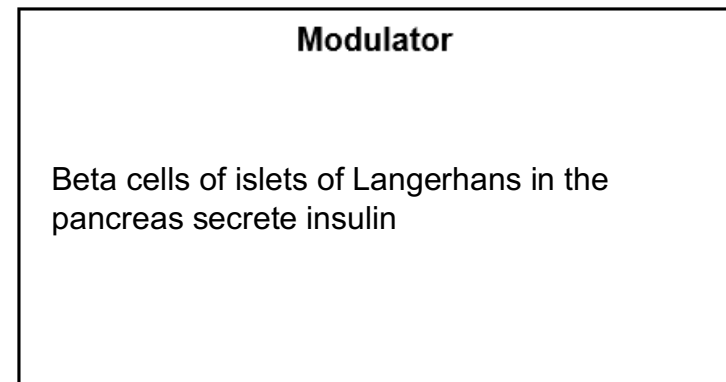
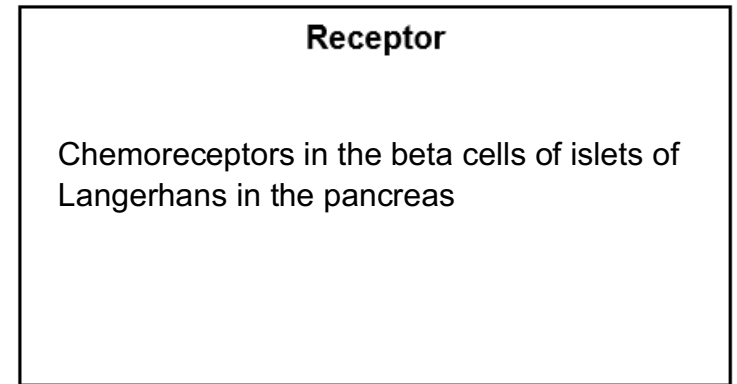
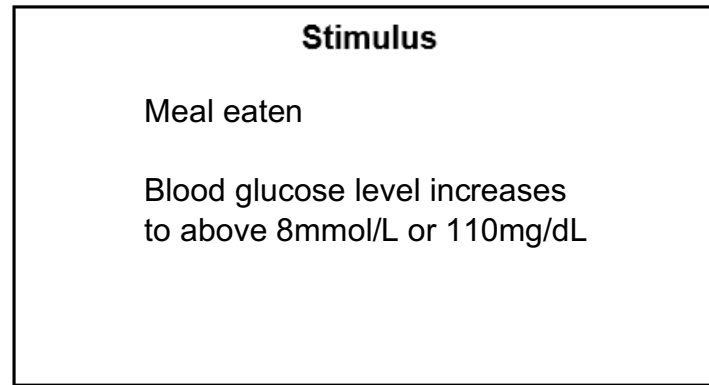
Gluconeogenesis: Producing glucose from fats and amino acids

Lipogenesis: Conversion of glucose to fatty acids for long term storage

Glycogenolysis: Process of converting glycogen back to glucose

Blood Glucose Levels

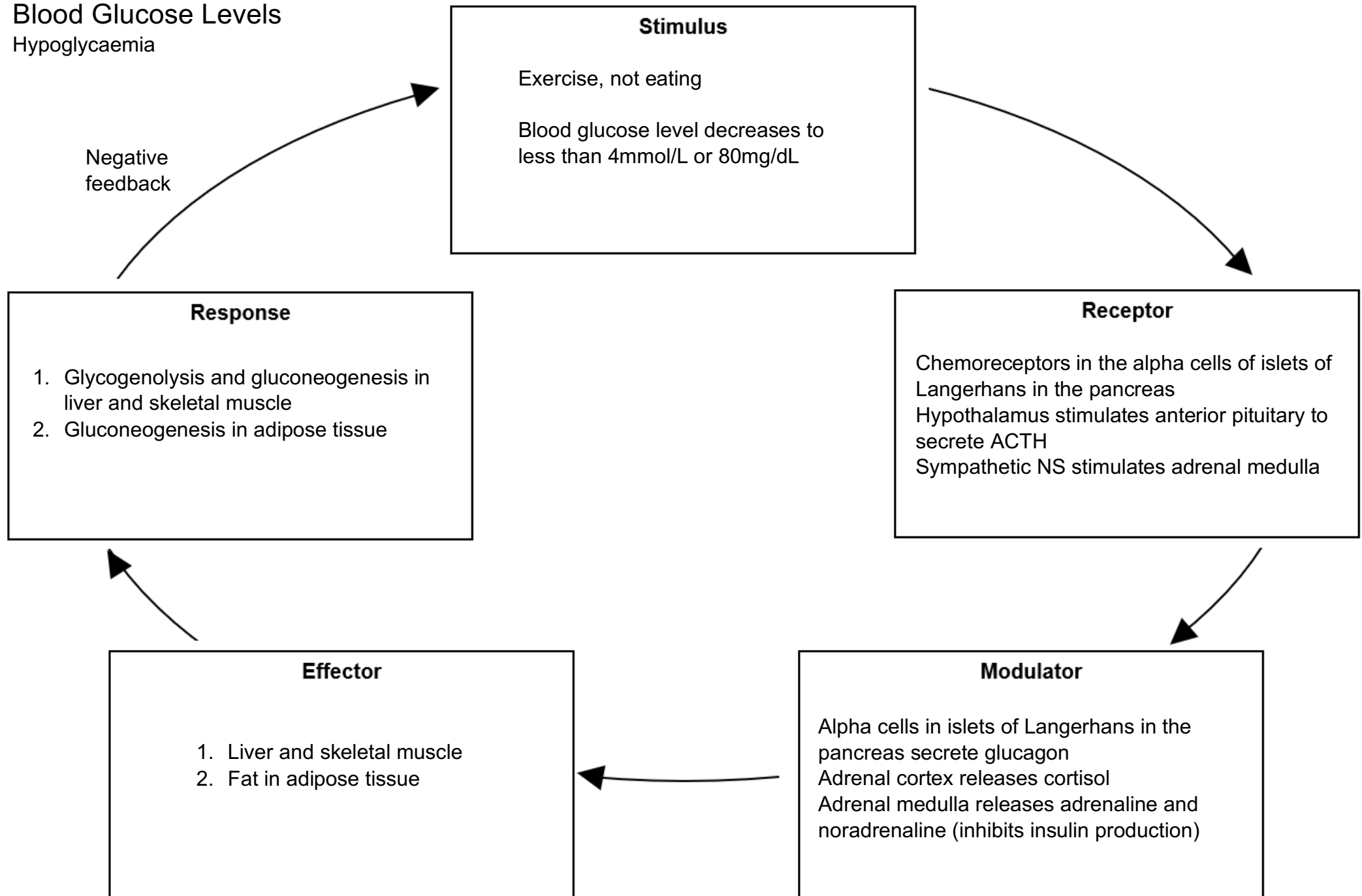
Hyperglycaemia



Negative feedback

Blood Glucose Levels

Hypoglycaemia



Osmoregulation

- Body approx. 60% water
- Types of fluid:
 - Intracellular: fluid inside cells
 - Blood plasma
 - Intercellular/extracellular/interstitial: fluid outside cells
- Water must be kept fairly constant and is obtained by drinking, eating and respiring
- 2.5L of water lost daily
 - lost through skin, lungs, kidneys and intestines

Kidneys

- 60% of water is excreted by the kidneys as it is the only water loss that can be regulated by the body
- Each kidney has approx. 1.2 million nephrons

Control of water loss

- Level of reabsorption of water is controlled by Anti-Diuretic Hormone (ADH)
 - Causes distal convoluted tubule (DCT) and collecting duct to reabsorb more water
 - Secreted by posterior pituitary
- When we are short of water, osmotic pressure rises
 - Water from interstitial fluid diffuses into plasma
 - Water from intracellular fluid diffuses across membranes into intracellular fluid
 - Cell shrinks

Dehydration

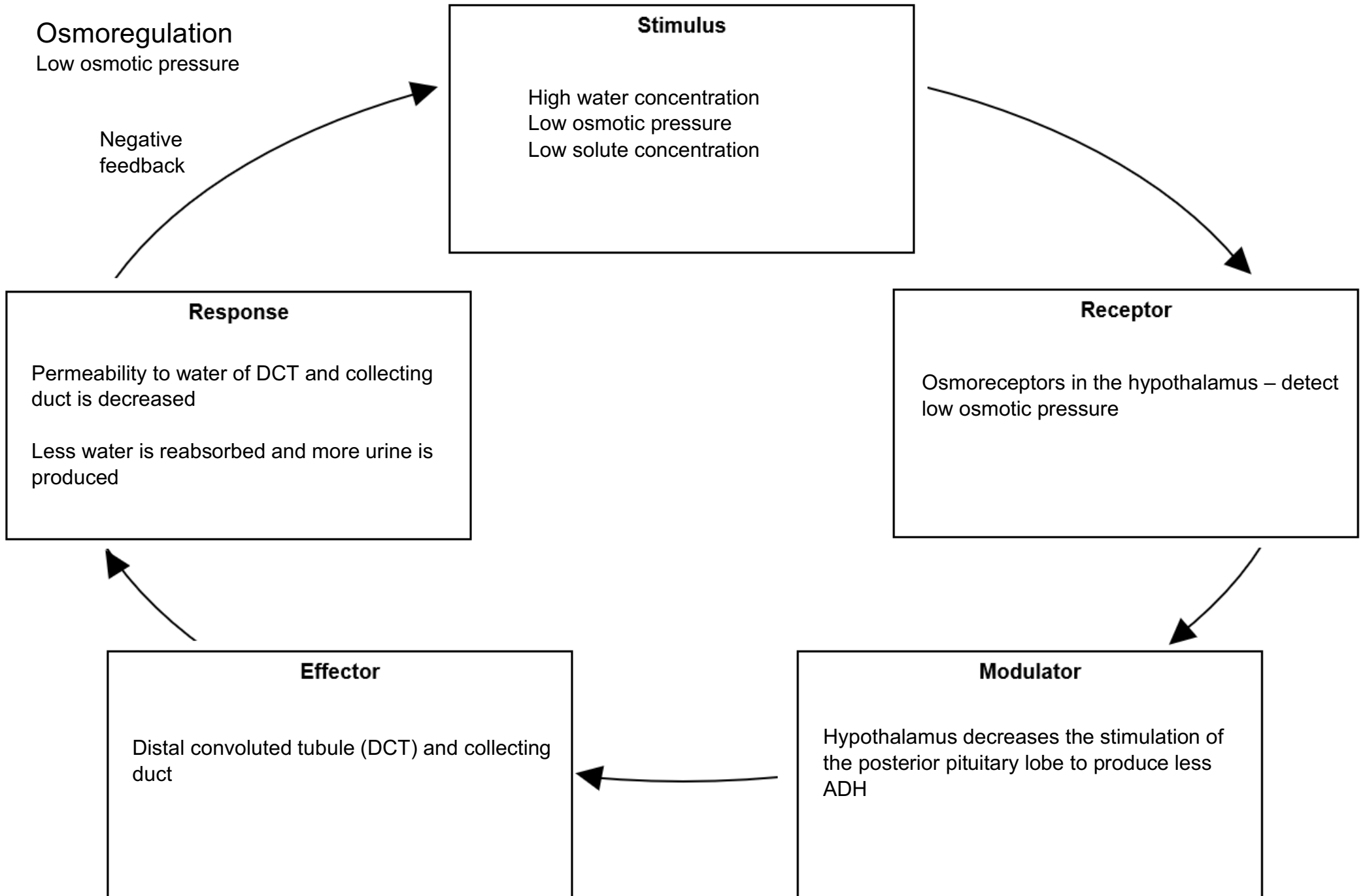
- Too little water
- Can be caused by sweating, vomiting, diarrhea
- Leads to headaches, dizziness, thirst, deliriousness and death

Intoxication

- Too much water
- Can be caused by drinking too much water
- Leads to headaches, vomiting, light-headedness, collapse, and rare: death

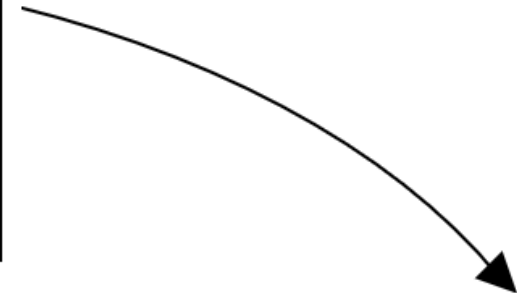
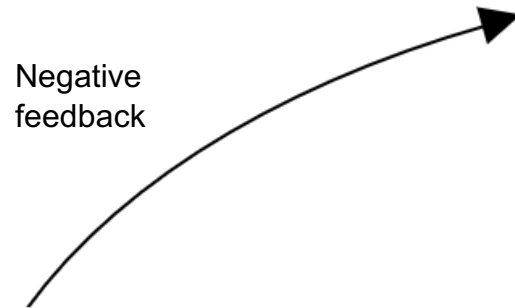
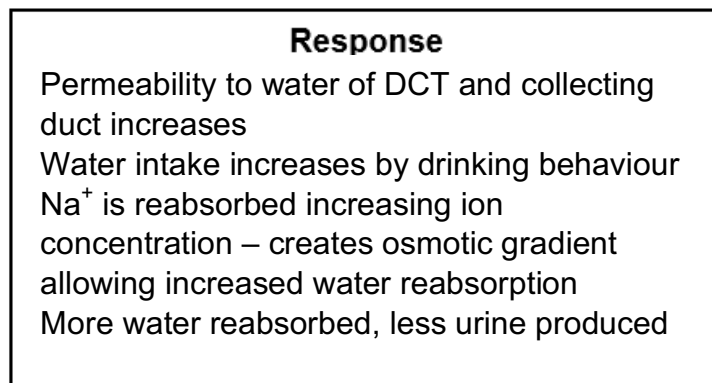
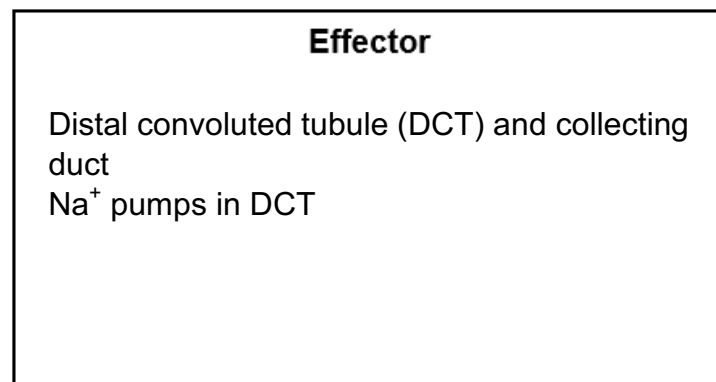
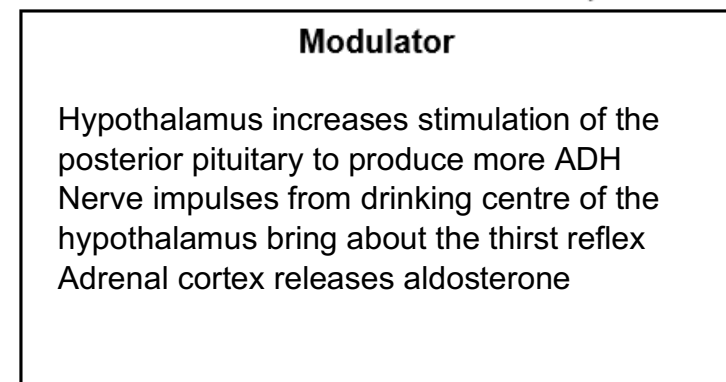
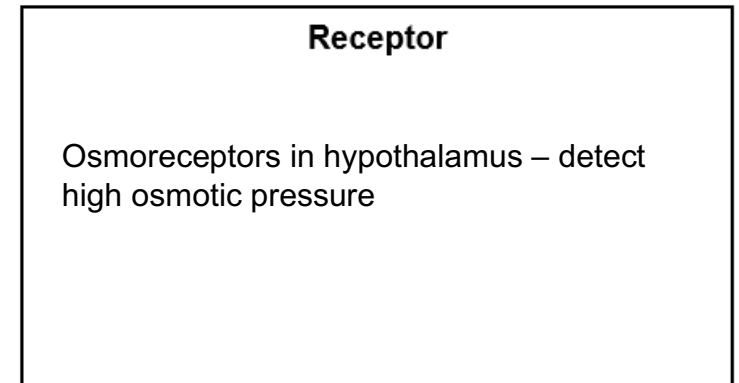
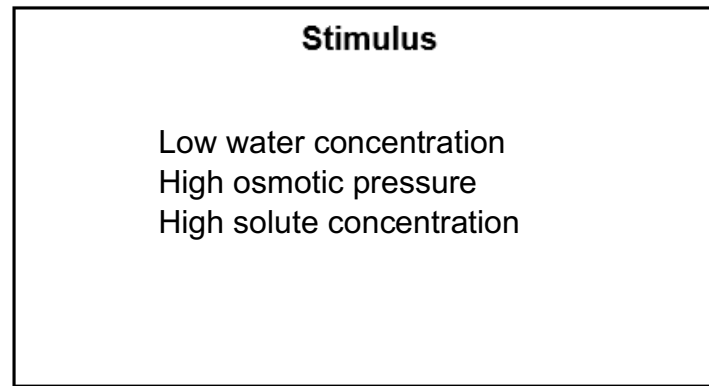
Osmoregulation

Low osmotic pressure



Osmoregulation

High osmotic pressure



Gas Concentrations

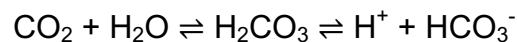
- All cells need a constant supply of oxygen
- Carbon dioxide must be removed from the blood stream
- Diaphragm and intercostal muscles carry out breathing
 - Stimulated by phrenic and intercostal nerves – controlled by medulla oblongata

Oxygen concentration

- O₂ does not affect rate of breathing unless it is at dangerously low levels
- Chemoreceptors in aortic and carotid bodies detect O₂ concentration in the blood
- Chemoreceptors also in medulla oblongata

Carbon dioxide concentration

- CO₂ the major stimulatory factor for breathing
- Chemoreceptors respond to increase in H⁺ ions and CO₂ concentrations



- More H⁺ ions – lower blood pH

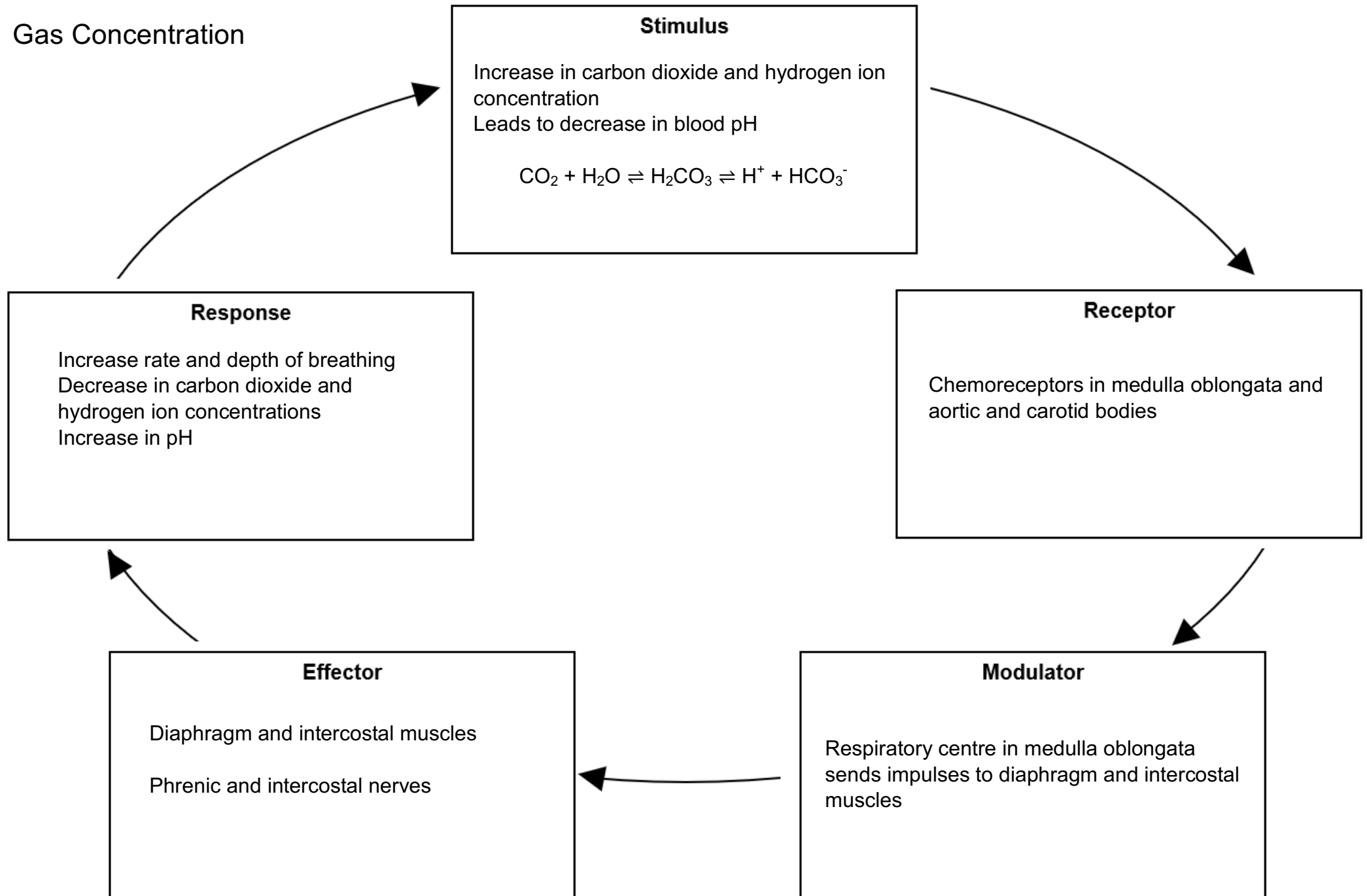
Voluntary control of breathing

- Needed for speech, music, swimming, etc.
- Build up of CO₂ forces us to take a breath

Hyperventilation

- Rapid deep breathing that provides more oxygen and less carbon dioxide than what is necessary for the body
- Hyperventilation before swimming means that the swimmer may lose consciousness due to the lack of oxygen before needing to take a breath due to carbon dioxide build up
 - Can lead to drowning

Gas Concentration



Response to Infection

Pathogen: A disease-causing organism; often referred to as a pathogenic organism

Types of Pathogens

Bacteria

- Mostly non-pathogenic
- Lots on our skin and in our body
- Uni-cellular shape



Rod-shaped
Bacillus



Spherical
Cocci



Spiral
Spirillum



Curved rod
Vibrio

Viruses

- When a virus infects a living cell its DNA or RNA induces the cell to manufacture more virus particles
- Some viruses multiply in bacterial cells – these are called bacteriophages
- Virus infected cells secrete cytokines called interferons, which introduce resistance to viral infection in the surrounding cells

Transmission

- Contact
- Body fluids
- Droplets
- Ingestion
- Airborne

Examples

- Chlamydia
- Cholera
- Diphtheria

Transmission

- Contact
- Body fluids

Examples

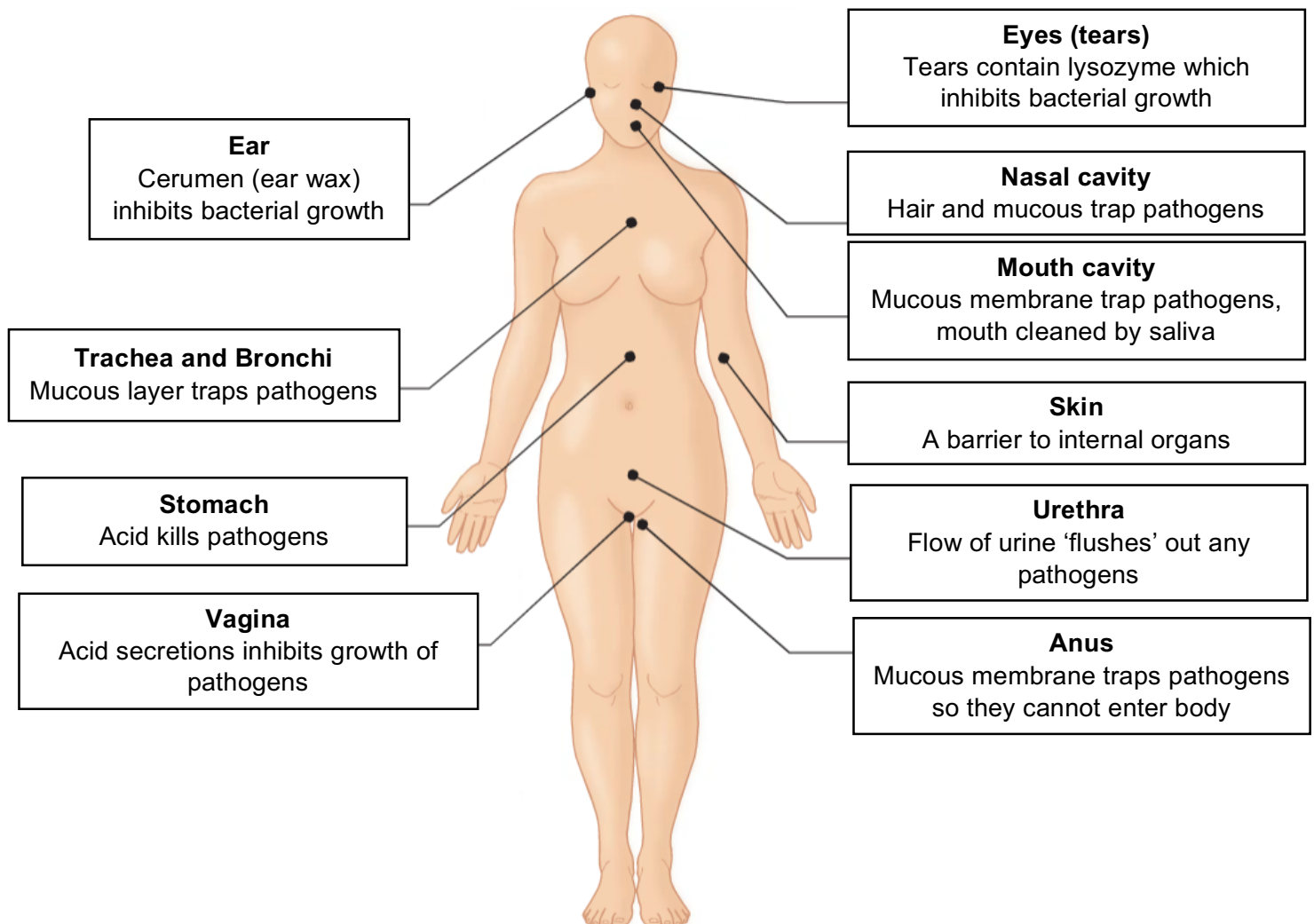
- HIV/AIDS
- Influenza
- Measles

Transmission of Pathogens

Contact	• Spread of pathogen by physical contact • May be direct: actually touching an infected person • May be indirect: touching an object that has been touched by an infected individual • Skin infections and STIs
Body fluids	• Transfer of body fluids from one person to another • Blood or other body fluids from an infected person comes into contact with the mucous membranes of an uninfected person, then pathogens may enter the body of that person • HIV, Hep B and C
Vectors	• Transfer of pathogens by other animals (insects, ticks, mites) • Some vectors transfer the pathogen directly • Many vector-borne diseases are spread by a specific vector • Malaria, dengue fever, lyme disease, bubonic plague
Contaminated food and water	• Contamination of food and water during food preparation, or through vectors that spread the pathogen to food or water • Cholera, Hep A, Typhoid

First Line of Defence

- External
- Non-specific



- External
- Non-specific

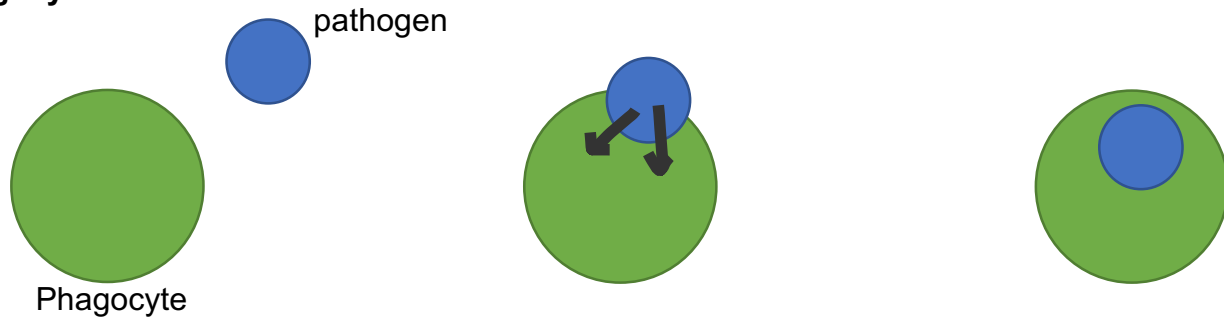
Protective reflexes

Sneezing	Stimulus: irritation of the nasal cavity Response: Forceful expulsion of air from lungs results in mucous and foreign particles to be expelled through nose and mouth
Coughing	Stimulus: Irritation of lower respiratory tract Response: Air is forced from the lungs to try and release or remove the irritant. Can release mucous and foreign particles out of the trachea and into the throat and mouth
Vomiting	Stimulus: As a result of bacterial toxins or due to overstretching (eat too much) Response: Involuntary forceful expulsion of stomach contents
Diarrhoea	Stimulus: Irritation of the walls of the small and large intestine Response: Faeces is more watery leading to faster expulsion of foreign bacteria/viruses from the anus

Second Line of Defence

- Internal
- Non-specific

Phagocytosis



1. Phagocyte attracted to the foreign particle. Ingestion by the phagocyte/macrophage occurs
2. Vacuole forms inside the phagocytic cell. A lysosome binds to the vacuole and releases digestive enzymes
3. Enzymes break down the foreign particle leaving soluble debris
4. Vacuole leaves via exocytosis

Inflammation caused by any injured cell

1. Tissue is damaged. The skin becomes red, swollen, hot and painful
 2. Mast cells (in fluids) stimulate production of histamine, heparin and phagocytes
- Histamine: increase blood flow, causing capillaries to become more permeable to blood – increases phagocytes to the area. Increase metabolic heat (causing redness) with more fluid being filtered, the site swells.

Heparin: prevents blood clotting. Allows phagocytes to be attracted to chemicals released by the mast cells. Blood clots form around injury, slowing down blood flow and pathogen movement. Only occurs for short time before phagocytic activity occurs

Phagocytes: are attracted to chemicals released by the mast cells

3. Pain receptors are stimulated, blood clotting occurs
4. The phagocytic cells fill with debris, forming pus

Fever

- Increase in body temperature to above 37.5°C – usually 39°C
- As macrophages attack a pathogen, they secrete interleukins/pyrogens into the blood
- Pyrogens/chemicals work directly on the hypothalamus, triggers an increase in body temperature
- Body feels cold and responds by shivering and vasoconstriction in skin
- Body temperature falls to normal

Benefits of fever:

- A higher temperature is thought to prevent the growth of some bacteria and viruses
- Increases the effects of interferon (introduce resistance to viral infection in surrounding cells and prevent replication of virus particles in host cells)

Third Line of Defence

- Internal
- Specific

Antibody-mediated / humoral response

Antigen: any substance capable of causing formation of antibodies when introduced into the tissues

Self-antigen: any large molecule produced in a person's own body; does not cause an immune response in that person

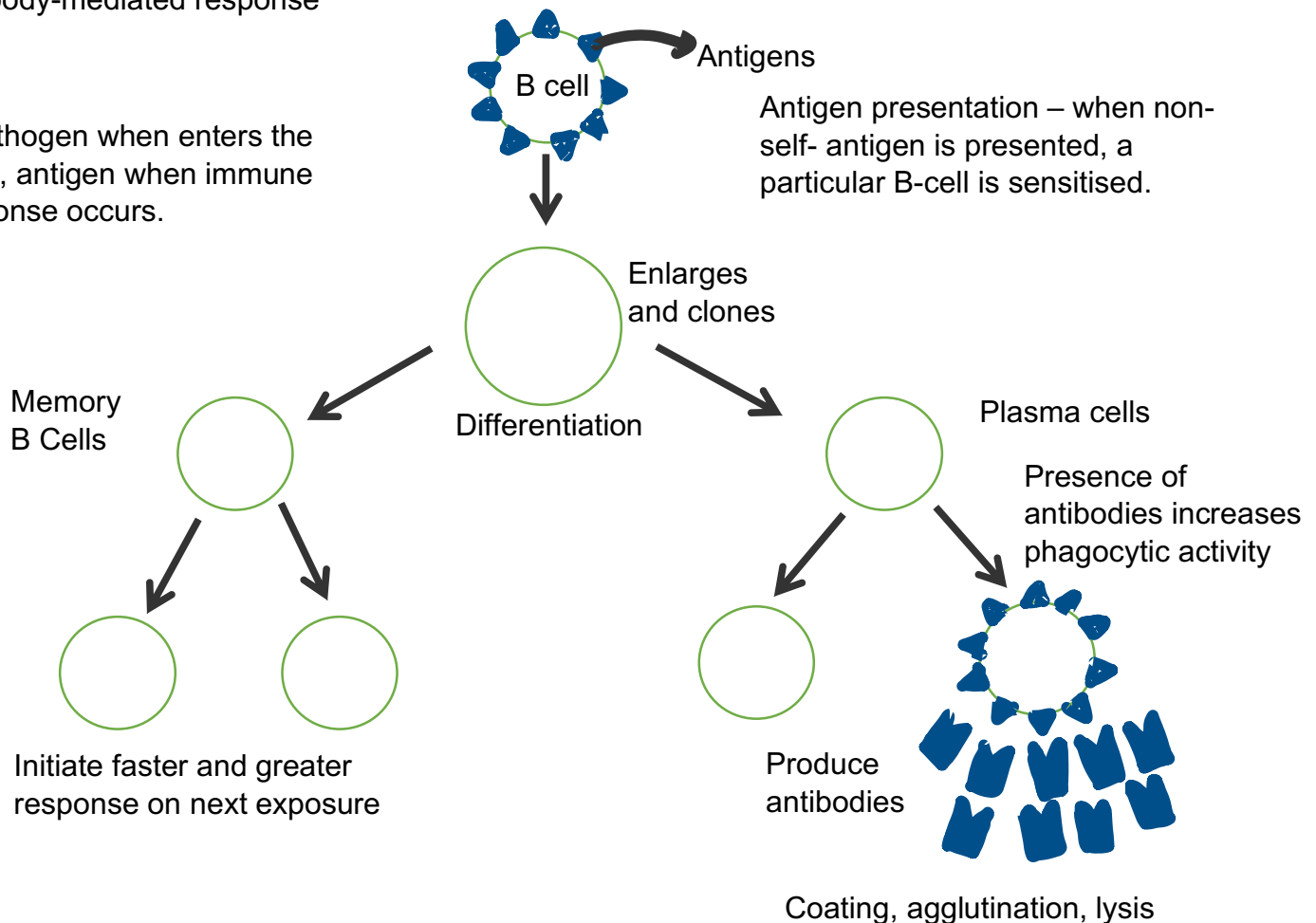
Non-self-antigen: any compound foreign to the body that triggers an immune response

Antibody: a substance produced in response to a specific antigen; combines with antigen to neutralise or destroy it

- B-lymphocytes/B-cells
- Gives resistance to intercellular/extracellular antigens
- Specific antigen engulfed by macrophages
- Antigens presented to specific B cells in lymphoid tissues
- Each B cell has a specific receptor protein that match a specific antigen (antigen-antibody complex)
- B cell is sensitised, enlarges, clones, divides and differentiates into:
 - Memory cells for faster and larger response at next exposure to the pathogen
 - Plasma cells that make antibodies specific to the antigen and release into lymph to attack antigens
- Antibody action: inactivate, dissolve, lysis, clump (agglutinate), coat to attract macrophages

Antibody-mediated response

A pathogen when enters the body, antigen when immune response occurs.



Antibody action

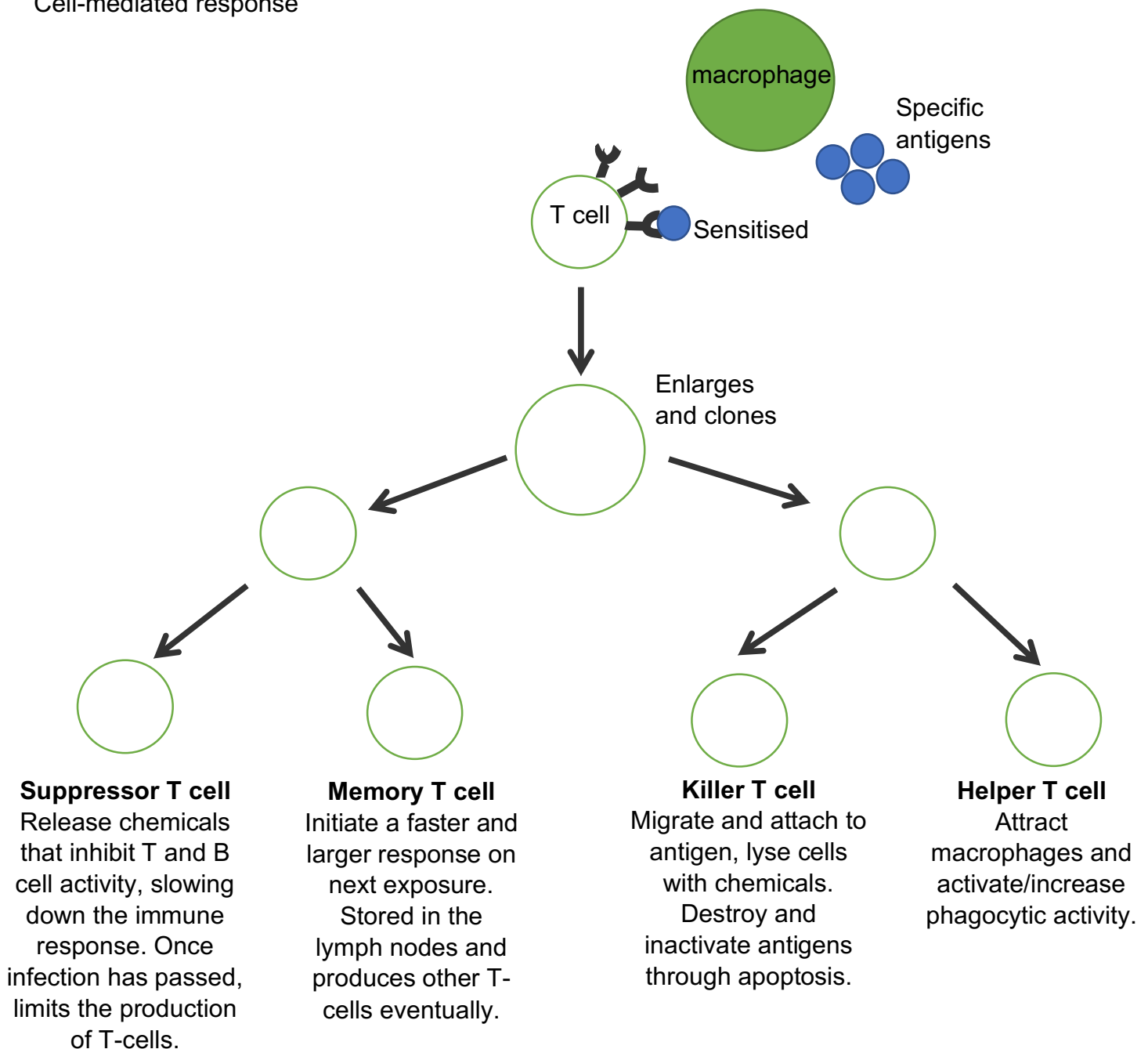
Antibodies combine with the antigen to form an antigen-antibody complex

- Coating - pathogen to attract phagocytes and digest easier
- Agglutination – pathogens clump together to stop them spreading
- Lysing – punch holes in bacterial cells so they burst/lyse
- Binding – bind to pathogen surface, stops them entering or damaging cells
- Dissolving – dissolve the pathogens

Cell-mediated response

- Attack pathogens that get inside cells (intracellular) where antibodies are unable to get to the pathogen, e.g. viruses
- T cell receptors are activated by antigen-presenting macrophages or B-cells
- Mobile, move around the body, mature in the Thymus
- Stimulate B cells and macrophages that lyse/kill cells e.g. cancer and transplant cells

Cell-mediated response



Types of Immunity

	Natural No human intervention	Artificial Injection, etc. of antibodies
Active Long lasting immunity	Antigens enter the body naturally, microbes cause the person to catch the disease. Stimulates the immune response but suffers symptoms. Body produces own lymphocytes and antibodies	Immunisation/vaccinations. Antigens (weakened or dead microbes or subunits) are injected as vaccines. Immunity without symptoms – body produces antibodies and specialised lymphocytes
Passive Short lasting immunity	Antibodies pass from mother to foetus via placenta during pregnancy or via breast milk during breast feeding. A baby does not produce antibodies of its own	Anti-venom. Pre-formed animal antibodies are injected into the body (treat snake bites, tetanus). The body does not produce antibodies

Treatment of Infection

Antibiotics

- Used for fighting the infection of bacteria
- Two types:
 - Bactericidal antibiotics: kill bacteria by changing the structure of the cell wall or cell membrane or disrupting the action of essential enzymes
 - Bacteriostatic antibiotics: stop bacteria from reproducing, usually by disrupting protein synthesis
- Broad spectrum: affect a wide range of bacterial infections
- Narrow spectrum: effective against only a couple of types of bacterial infections
- Multiple drug resistance: some strains of bacteria are resistant to most antibiotics, hastened by the overuse of antibiotics
- Total drug resistance: the resistance of some strains of bacteria to all antibiotics

Antivirals

- Antibiotics do not affect viruses
- Antivirals specific to viruses
- Inhibit the development of the virus

Vaccines

- 4 types of vaccines

Type	Explanation	Examples
Living attenuated micro-organisms	Micro-organisms of reduced virulence (disease-producing power)	Measles, mumps, rubella, TB
Dead micro-organisms	Micro-organisms that have been cultured and killed	Cholera, typhoid, whooping cough, influenza, Hep A
Toxoids	Filtrates of bacterial cultures containing inactivated toxins	Diphtheria, tetanus
Sub-unit	A fragment of the organism is used to make the vaccine	HPV, Hep B

Risks of Vaccines

- Allergic to the culture medium, e.g. Influenza vaccine cultured in egg protein
- Inability to completely isolate a virus from another, risk of introducing cross species virus – recombinant DNA prevents this
- Vaccines use preservatives such as Formaldehydes and acetone which some claim affect use of nervous system

Ethical Concerns

- Ethical concerns over vaccines centre around its manufacture, testing and risks
- Other vaccines require human tissue to manufacture, this eliminates risk of cross species viruses being introduced
- Rubella vaccine is manufactured using cells of aborted foetuses which is an ethical concern for those who oppose abortion, they must weigh up health vs. stance on abortion
- Trial groups for vaccines may not be aware of the risks involved
- Vaccines usually tested on animals before human trials begin
- Some country's governments do not subsidise vaccines – economical concerns for families

Participation in vaccination programs

- The decision to participate in vaccination programs may be influenced by social, economic or cultural background
- In Australia, most vaccination programs are subsidised by the government. Majority of Australians that choose not to vaccinate are doing so for social or cultural reasons
- Religion may also be a factor – this coincides with ethical concerns.
- Cultural influences – traditional medicine is considered 'superior' to the evidence-based medicine practised in most developed countries.

UNIT 4

Mutations

Term	Definition
Allele	An alternative form of a gene; an individual normally has only two alleles of each gene
Allele Frequency	How often an allele occurs in the gene pool
Chromosome mutation	A change to the structure and/or number of chromosomes in an organism
Founder effect	A type of genetic drift that occurs when a new population is formed by a small number of individuals; the small size of the sample can cause marked deviations in allele frequencies from the original population
Gene	The factor that determines hereditary characteristics
Gene flow	The transfer of alleles from one population to another through migration
Gene mutation or point mutation	An alteration to a single gene
Gene pool	The sum of all the alleles carried by the members of a population
Genotype	
Germ line mutation	Mutation affects the reproductive cells/gametes. It does not affect the individual, but the next and subsequent generations. May result in characteristics that increase chances of survival – these characteristics are likely to be passed on to another generation, resulting in a change of the allele frequency of gene pools.
Lethal recessive	
Mutagen	An environmental agent that increases the rate of mutation
Mutation	A change in a gene or chromosome leading to new characteristics in an organism
Natural Selection	Established by Darwin • Variation – there are differences within a population. Some differences may give a selective advantage • Over production – more individuals are produced than can be sustained by the environment • Struggle for existence – due to excessive birth rates and limited resources there is a struggle between individuals • Survival of the fittest – only the best adapted individuals survive and reproduce • Like produces like – the favourable characteristics are passed on to the next generation • Over time the gene pool changes – the proportion of favourable alleles increases and less favourable decreases
Phenotype	
Population	A group of organisms of the same species living together in a particular place at a particular time
Random genetic drift	The change in allele frequency in a population as a result of chance rather than natural selection; occurs only in small populations
Selective advantage	
Selection pressures	The extent to which organisms possessing a particular characteristic are either eliminated or favoured by environmental demands. It indicates the degree of intensity of natural selection.
Somatic mutation	Mutation occurs in the body cells. The characteristic of the somatic mutation only affects the individual – i.e. the mutation is lost when the individual dies. Somatic mutations often cause cancerous growths

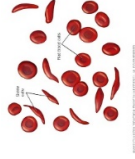
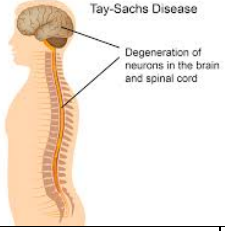
Speciation	Not a mechanism for evolution as it is the result of evolutionary mechanisms at work. The process by which new species are formed from existing ones: 1. Variation within a population 2. A barrier results in a separation of the population. Two groups initially sharing the gene pool will eventually have separate gene pools 3. The environmental differences provide different selection pressures resulting in a change in gene frequency within the 2 separate gene pools 4. Over time, subspecies develop which are isolated but could still interbreed to provide fertile offspring 5. Over a prolonged period the changes in gene frequency make it impossible for the 2 groups to interbreed and produce fertile offspring. 2 new species form
Species	The basic unit of biological classification; members of a species are capable of interbreeding and producing fertile offspring
Variation	The differences that exist between individuals or populations of a species

Gene Mutations

- Small, localised change in the structure of the DNA strand
- May alter triplet codon reading
- Amino acids changed
- Most are silent, i.e. do not express as the same amino acid is joined to the chain
- Three types:

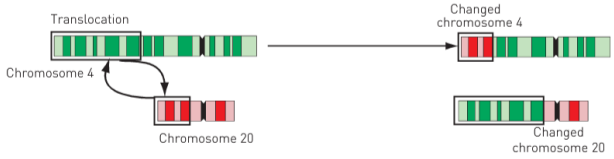
Type	Diagram	Explanation
Substitution	 TGGCAG ↓ TGGTAG	• Single base substituted for another base • May result in a different amino acid, or a stop codon
Insertion	 TGGCAG TGGTATCAG	• Extra base added to the DNA strand • Frameshift right, completely new sequence of amino acids
Deletion	 TGGCAG TGGG	• Base removed from the DNA strand • Frameshift left, completely new sequence of amino acids

Case Studies

Genetic Disease	Population	Cause	Main symptoms	Inheritance/gene pool
Sickle-cell anaemia 	Black African	Substitution of A to T in the beta chain for haemoglobin on Chromosome 11	- Sickle red blood cells - Swelling of hands and feet - Fatigue - Jaundice	Recessive trait passed on from affected parents Selective advantage for malaria – plasmodium cannot easily invade sickled cells Homozygous recessive dies early, homozygous dominant susceptible to malaria Heterozygous more likely to survive malaria and reproduce, increase sickle-cell allele frequency
Tay-Sachs disease 	Ashkenazi Jews of Eastern Europe Incidence 1 in 2500 live births	Defect in coding for Hex A enzyme for fat metabolism on chromosome 15	- Build up of fatty substances in nervous system - Seizures - Blindness - Deafness - Reduces ability to swallow	Recessive trait passed on from affected parents – barrier to gene flow as many marry within their faith Homozygous recessive individuals die early Heterozygous form provides selective advantage for tuberculosis Heterozygous more likely to survive TB and reproduce, increase allele frequency
Thalassemia 	People of Mediterranean descent	Alteration in gene that produces alpha or beta haemoglobin chain on chromosome 11 More mutations in red blood cell formation increases severity of disease	- Defects in formation of haemoglobin - Anaemia - Iron deficiency - Slow growth - Require frequent blood transfusions and chelation therapy - May have bone marrow transplant	Recessive trait passed on from affected parents – barrier to gene flow as normal for cousins to marry each other. Small gene pool. More mutations in gene pool means a higher mortality rate

Chromosome Mutations

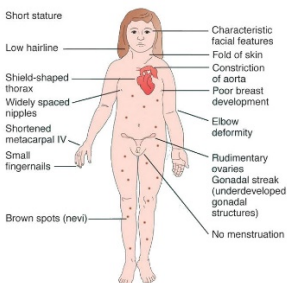
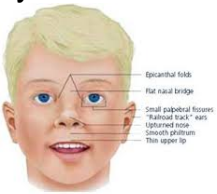
- Occur when more than one gene is affected and may result in a mutation in all or part of a chromosome

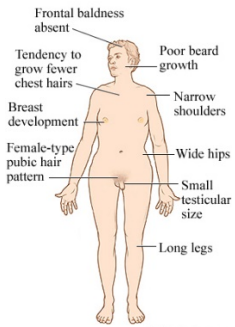
Duplication Section of chromosome occurs twice. Chromatid may break off and join the wrong chromatid 	Deletion Part of the chromosome is lost 
Inversion Broken piece of chromosome joins the wrong way around. May disrupt pairing of homologous chromosomes 	Translocation Part of a chromosome breaks off and joins to the wrong chromosome 

Non-Disjunction

- Occurs in meiosis II
- Sister chromatids fail to separate properly during anaphase II
- One gamete receives an extra chromosome and one has one
- Some gametes are unaffected

Case Studies

Genetic disease	Cause	Main symptoms
Monosomy		
Turner syndrome 	- Females have a missing X chromosome - XO genotype	- Treatment aimed at treating physical defects - Mental development normal but difficulty with spatial memory - Low levels of reproductive hormones, results in low development of secondary sexual characteristics and degenerate ovaries, almost always infertile - Reduced stature, body typically short
Cri-du-chat (cat cry) syndrome 	Part of chromosome 5 is missing	- Infants often have a high-pitched cry that sounds like a cat - Intellectual disability and delayed development - Small head size - Low birth weight - Weak muscle tone - Distinctive facial features – low set ears and a rounded face

Trisomy		
Klinefelter Syndrome 	- Males have an extra X chromosome - XXY or XYY genotype	- Low intelligence - Low testosterone levels - Underdeveloped penis and testes - Some breast development
Down Syndrome 	- Chromosome 21 has an extra chromosome 1 in 800 births	- Slanting eyes - Abnormal ears - Poor muscle tone
Patau Syndrome 	- Chromosome 13 has an extra chromosome 1 in 3000 births	- Polydactyly (extra finger/toe) - Cleft palate - Structural eye defects

Mechanisms of Human Evolution

- Information is passed on from one generation to the next (inheritance)
- DNA codes for polypeptides which in turn codes for proteins or enzymes. This section of DNA is a gene

Evolutionary Mechanisms

- Even within a species there are variations, brought about by:
 - Crossing over during prophase I in meiosis
 - Independent assortment of chromosomes during metaphase I and II
 - Random mate choice and fusion of gametes
 - Gene and chromosome mutations
- 4 factors affect allele frequency: natural selection, random genetic drift, founder effect and migration

Mechanism	Explanation
Natural selection	• Process by which a species becomes better adapted to its environment • Those individuals with favourable characteristics have a survival advantage and pass those characteristics onto subsequent individuals • Not random • Alleles passed on tend to be the ones that enhance survival Mechanisms of natural selection • Variation – there are differences within a population. Some differences may give a selective advantage • High birth rate – more individuals are produced than can be sustained by the environment • Struggle for existence – due to excessive birth rates and limited resources there is a struggle between individuals • Survival of the fittest – only the best adapted individuals survive and reproduce • Like produces like – the favourable characteristics are passed on to the next generation • Over time the gene pool changes – the proportion of favourable alleles increases and less favourable decreases
Random Genetic Drift	• Also called Sewall-Wright effect • Often a random, non-directional variation in allele frequencies Case Study: Dunkers from Pennsylvania. Did not marry outside their own religious community. Became an isolated group within the US Case Study: Bentinck Island in Aus. Originally part of the mainland but rising sea levels cut off the island. No I^A allele present, high I^B allele, as by chance there were no I^A genotypes • These small isolated communities exist within a larger population, so changes in allele frequency within these gene pools are as a result of random changes between phenotypes within a population
Founder Effect	• Occurs when a small group moves away from its homeland to a new area and establishes a community which later expands • The small size of group can cause marked deviations in allele frequencies from original population Case Study: Afrikaner population of Dutch settlers in South Africa which is descended from a few settlers. Today the Afrikaner population has an unusually high frequency of the gene that causes Huntington's disease Case Study: Pingelap population of Micronesia. A typhoon reduced the population to 20. High frequency of the achromatopsia (colour blindness) allele
Migration	• Gene flow from one population to another • If immigrants from one country bring alleles that are not already in the population, the frequency of the alleles of that gene will be altered

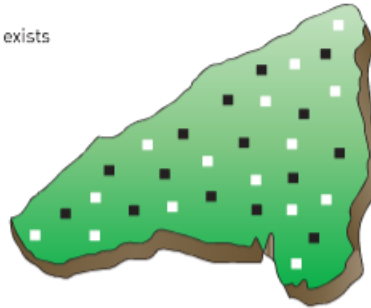
Factors Affecting Gene Flow

- Geographical barriers
 - Oceans, mountains, etc.
- Sociocultural barriers
 - Religion: marry within religious groups
 - Language: can't talk to each other
 - Socioeconomic status

Speciation

a Variation

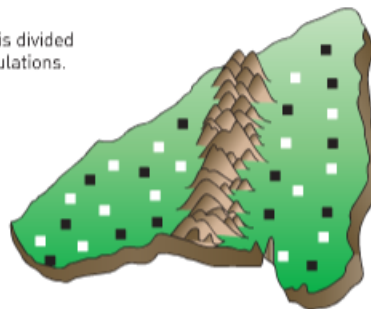
A population exists on an island.



A range of **variations** exists within the population, which shares a **common gene pool**.

b Isolation

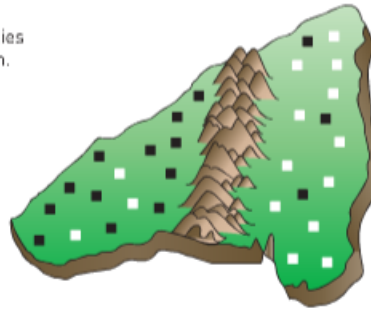
The species is divided into two populations.



A **barrier** has formed, dividing the population into two. No interbreeding occurs between the two populations. Each population has a **separate gene pool**.

c Selection

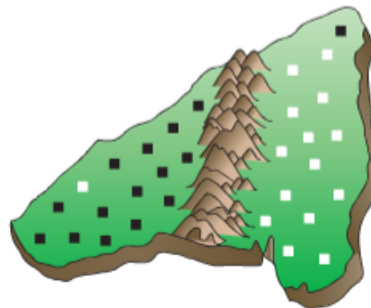
Two subspecies begin to form.



Different **selection pressures** act on each of the two populations over a number of generations. This brings about a change in the **gene frequencies** of each gene pool. Such changes lead to the evolution of separate **subspecies**.

d Speciation

Two species now exist.



Over a long period of time the changes in the gene frequencies may be great enough to prevent the production of fertile offspring by interbreeding between the two populations from ever occurring again. When this happens, **two species** exist.

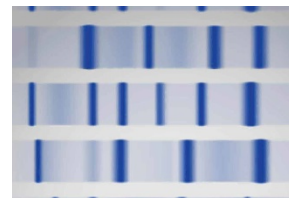
Biotechnology

- Used to know where faulty genes are located and the possibility of replacing these genes through gene replacement
- Some genes which are responsible for generic disease are expressed later in life, knowing we have these genes will allow us to take preventative measures like alcohol and diet
- Biotechnology has lead to mutations being identified and linked to diseases

Gel electrophoresis

- When a current is run through the gel, the negatively charged pieces move towards the positive electrode, the smaller pieces move faster than the larger ones
- Results in a unique banding pattern
- Known as DNA profile and can be used to identify people

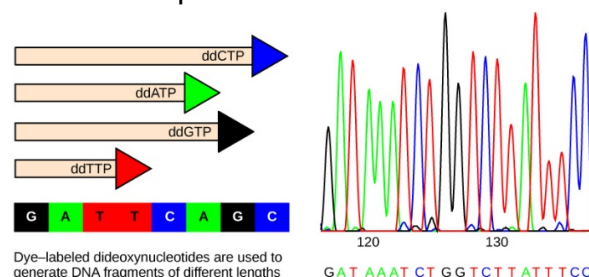
- (1) Enzymes are used to cut DNA at specific base sequences
- (2) DNA pieces placed on bed of semi-solid gel
- (3) Electric current passed through electrodes located at either end
- (4) The DNA (negatively charged) moves towards the positive electrode
- (5) The smaller DNA pieces move faster than the larger pieces resulting in a pattern of bands



DNA Sequencing (Sanger sequencing)

- Precise order of nucleotides
- Done by introducing synthetic nucleotides to a growing strand. Synthetic nucleotides do not contain a hydroxyl (OH) group essential for the next nucleotide to attach to. Stops elongation of the sequence
- By comparing strands which have been stopped mid-replication we can detect changed alleles
- Knowing the location of the gene/allele allows sequencing to be performed and determination of a faulty allele. Individuals can then seek effective or preventative treatment

- (1) Make more DNA for sequencing
- (2) Use heat to denature (separate) the DNA
- (3) Place complementary DNA strands into 4 different reaction vessels
- (4) DNA polymerase added into reaction vessels. The DNA polymerase adds the nucleotides together
- (5) Free nucleotides added into reaction vessels. Each vessel has a different 'dye' for a specific nucleotide
- (6) Once nucleotide is base paired, the sequence is terminated because the nucleotide lacks a hydroxyl group
- (7) Gel electrophoresis is used to sequence the DNA



Polymerase Chain Reaction (PCR)

- Technique where a DNA segment is artificially duplicated many times using an enzyme called DNA polymerase
- Duplication initiated using a primer which is a DNA strand complimentary to the required DNA strand
- Results in a doubling of the required strand each time, quickly resulting in millions of the required DNA strand which can then be used to sequence and identify mutations

(1) Denaturing (96°C)

Heat the reaction strongly to separate, or denature, the DNA strand. At 96°C, human DNA polymerase is destroyed, so a more heat stable one is used, Taq (thermus aquaticus) polymerase.

(2) Annealing (55-65°C)

Once DNA has been coded, a primer is attached to each strand. This is a section of DNA with a known sequence, which attaches to a specific part of the DNA chain.

(3) Extension (72°C)

PCR only amplifies a certain region of DNA. Primers act as a point for the Taq polymerase to start adding complementary DNA nucleotides one at a time in a single direction to the single stranded DNA. This takes place at 72°C in a thermocycler and doubles the number of DNA strands. This process is repeated until enough DNA is produced for the tests required.

Gene therapy

- Treat or cure genetic abnormalities by replacing faulty genes with healthy, functioning genes
- Can be delivered by:
 - Vector (viruses)
 - Capsules and nose drops
 - Aerosol delivery through a nebuliser

Cystic fibrosis

- Thick mucous secreted by mucous glands. May clog air passages and trap bacteria, making a person with CF susceptible to infection
- Discovering Cystic Fibrosis Transmembrane Regulator (CFTR) helped to develop treatment for CF
- Scientists successfully corrected for faulty CFTR genes in cultured cells by adding normal copies of the genes to the culture
- Delivered by modified common cold virus
- Gene therapy for CF may need to be given repeatedly, as gene expression does not last long. This is due to epithelial cells of respiratory tract being constantly replaced

Huntington's disease

- Gene therapy can be used to slow down or prevent progression of disease
- Caused by a mutation in gene on chromosome 4, called IT15
- Mutated form of protein huntingtin results in nerve cells in the brain being damaged. Causes physical, mental and emotional changes
- Researchers able to turn off/silence the gene, however neurons may not be able to survive without the protein
- Corrective gene has been experimented with using modified virus as a vector.

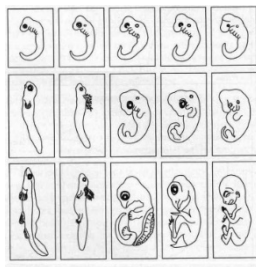
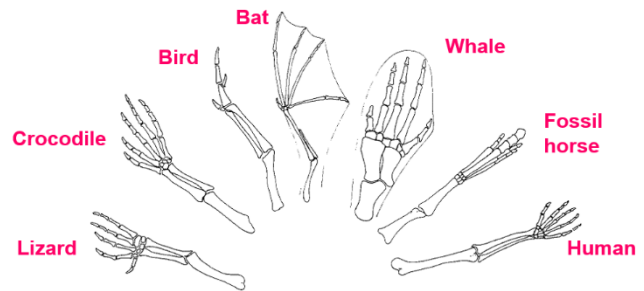
Evidence for Evolution

Evolution: the gradual process of change in the inherited traits of a population of organisms from one generation to the next. It is a permanent change in a population's gene pool from one generation to the next.

Comparative Anatomy

Homologous structures

- Structures that are similar e.g. pentadactyl (fore) limb
- Similar structures with different functions
- Evidence of common ancestry

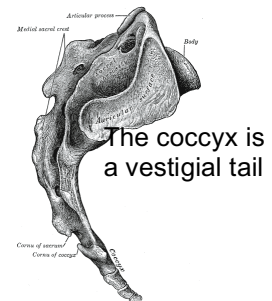
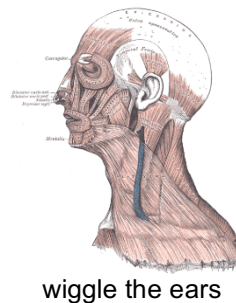


Embryology

- Many vertebrate embryos are similar
- Many have features not present in adults, e.g. gills, tails
- Common developmental pathway depends on closeness of relationship
- These provide evidence of common ancestry

Vestigial Structures

- Evolutionary left overs e.g. coccyx, appendix
- Structures that no longer serve any purpose
- Are reduced in size to conserve energy
- Provide evidence of change over time



Comparative Biochemistry

DNA Hybridisation

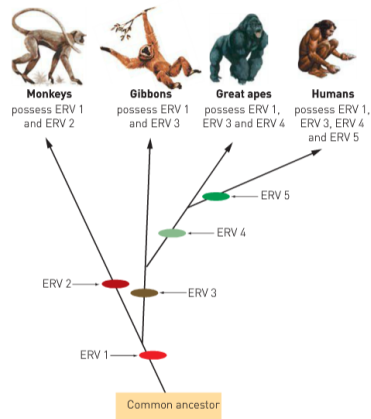
- Hybrid DNA is formed from different species
- More heat is required to separate strands that have been fused from 2 species that are most similar

ERVs

Endogenous retrovirus (ERV): A retrovirus that has become part of an organism's genome and exists in every cell of the body

- A virus is inserted into gamete DNA to compare non-functional DNA
- Upon entering a cell, a retrovirus copies its RNA genome into DNA through reverse transcription
- ERV's are passed on to all subsequent generations
- More closely related species have similar ERV's

Endogenous retroviruses



Mitochondrial DNA (mtDNA)

- Only inherited from the mother
- mtDNA has a higher rate of mutation
- It is estimated that the amount of mutation is proportional to the amount of time that has passed from the original female ancestor
- Able to estimate closeness of relationship through maternal ancestry
- Useful for same or closely related species
- Indicates more time has passed since common ancestry if there are less similarities

Protein Sequencing

- Sequences are very similar in closely related species and less similar in distantly related species
- Sequence similarities indicates evolution from a common ancestor
- Protein sequences for some proteins through a number of species has been completed, these are known as ubiquitous proteins
- Ubiquitous proteins perform basic but essential tasks required for life. The task is the same across all organisms
- Cytochrome C is a sequenced protein. Contains 104 amino acids, 37 of these have been found at the same positions in every sequenced Cytochrome C molecule
- This strongly suggests that these proteins have descended from a common ancestral Cytochrome C

Comparative Genomics

- Genome is a complete sequence of the base pairs that make up the entire DNA of an organism
- Comparative genomics allows comparison of two or more genomes
- Helps to identify preserved genes among species

Bioinformatics

- Base sequences in DNA can be expressed as data to be read by computer software
- IT platform for data provided by genomics – stores and manages the large amount of information provided by genomic studies
- Analysing and comparing genomes – can determine the evolutionary closeness of organisms

Fossil Evidence for Evolution

Fossils

Fossil: any preserved trace or remain left by an organism that lived in the past. E.g. footprints, burrows, impressions of all or part of an animal, bones, shells, teeth.

- Allows scientists to determine what extinct species were like, form an idea of what life was like in the past and build a sequence of evolution

Fossil formation

1. Rapid burial
Drifting sand, mud deposited by a river, volcanic ash or other members of the species.
Prevents or slows decay by microorganisms
2. Soil type
Must be wet, alkaline soil or sedimentary rock (new minerals). Bone minerals dissolve in acidic soil.
3. Presence of hard, bony parts
4. Absence of decay or organisms
5. Long period of stability (left undisturbed)

Human fossils

- Edges of rivers – sediment builds up after flooding or where water flow slows
- Caves – many contain limestone which deposits in dead organisms to preserve them or roofs/walls collapse and cover dead bodies
- Volcanically active areas – heat from lava usually destroys organisms but ash falls preserve many fossils

Fossil Dating

Absolute dating: dating methods that provide the actual age of the specimen in years

Potassium-Argon Dating

- Potassium-40 is radioactive and decays to form argon-40
- Decay occurs at an extremely slow but constant rate
- Half-life is 1.25 billion years.
- By determining the amounts of potassium-40 and argon-40 in a rock sample, the age of the rock can be calculated (not fossil)

Limitations

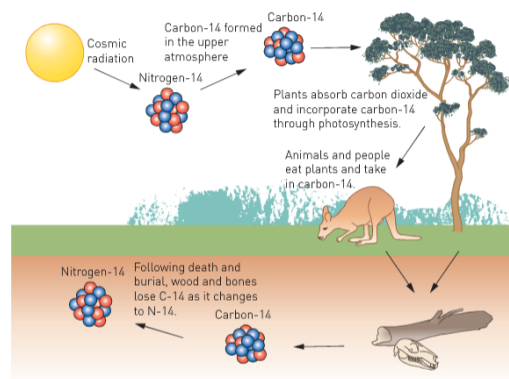
- Rocks must be at least 100 000 years old. If it is too recent, not enough potassium-40 will have decayed
- Only certain volcanic materials can be dated; there must be available rock of the same age as the fossil e.g. when rocks produced in volcanic eruptions bury bones.

Carbon-14/Radiocarbon dating

- Based on the decay of carbon-14, radioactive isotope of carbon to nitrogen
- Carbon-14 produced by the action of cosmic radiation on nitrogen in the upper atmosphere
- Plants absorb carbon dioxide and incorporate carbon-14 into their tissue through photosynthesis
- When animals and people eat plants, they take in carbon-14 and it is then incorporated into their tissue
- At death, the organism no longer takes in carbon-14, and the carbon-14 that is in their tissue starts to decay to carbon-12 and eventually nitrogen at a fixed rate. Carbon-14 has a half-life of 5730 years
- By measuring the ratio of carbon-14 to carbon-12, the age of the sample can be calculated

Limitations

- At least 3g of organic material need for radioactive decay to be measured
- Can only be used on samples less than 60 000 years old
- The material to be dated must be of organic origin (not rock)



Relative dating: dating methods that predict the age of the specimen in years

Stratigraphy

- Study of the layers (strata) of rocks
- Useful in 2 ways:

The principle of superposition

Assumes that in layers of sedimentary rock, the layers found at the top, and any fossils/artefacts within them, are younger than the material found lower down

Limitations

- The crust of the earth may have been distorted/disturbed by natural occurrences, leaving a sequence of rock layers turned upside down
- Fossils/artefacts could be buried by animals or early humans after the deposition of sediment; the specimen may be younger than some of the layers above.

Correlation of rock strata

Matching layers or rock strata from different areas. The rock itself is examined as well as the fossils in it. Rocks that contain the same fossils are assumed the same age.

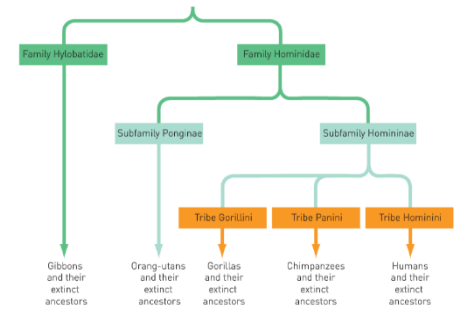
Index fossils: fossils widely distributed and present on earth for a limited time

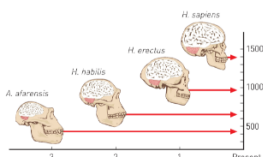
Problems with the fossil record

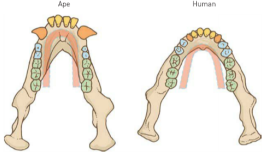
- Incomplete because of the specific conditions required for fossilisation – chance occurrence
- Must have wet, alkaline soil, rapid burial, absence of decay, left undisturbed, and presence of hard, bony parts

Hominid Evolutionary Trends

Hominids: classification is family. It includes all modern and extinct orangutans, gorillas and chimpanzees.



Characteristic		Trend
Skull	Size of brain 	- Increasing size relative to size of body. - Chimpanzees – cranial capacity of 393cm³ - Apes – cranial capacity of 400-500cm³ - Humans – cranial capacity of 1350cm³
	Convolutions in brain 	- The shape of the brains surface has been determined using endocasts. - Gradual increase in the number of convolutions in the surface of the cerebrum. - Greater development of frontal lobe, which is relatively small in apes.
	Cerebral Cortex	- Makes up an increasingly larger area of the brain. - Site of higher order processes/functions – vision, memory, reasoning, manipulative ability → develop skills. - Significant special skill brought about by cerebral cortex including tool making. - Changes in behaviour to meet a wide array of environmental problems.
Digits	Mobility 	- Pentadactyl (5 digits on each limb) - All digits can be moved independently
	Opposability 	1st digit (thumb) is opposable – increased effectiveness of opposability (tool use). There are two grips – power grip (hammer) and precision grip (holding a pen) - There is no opposability in feet due to upright bipedal stance. It has become a weight bearing appendage
	Claws/Nails 	- Hominids have nails. The order primates – some still have claws - Friction ridges (fingerprints) help increase grip between objects and digits. They have sense receptors on the ends of fingers to help grip and manipulate. - Two grips – power and precision - Old world monkeys have less manipulative ability
	Prehensile	Capable of grasping and it is essential for climbing allowing the ability to wrap digits around branches of trees.
Dentition	Dental Formula No. of each type of teeth in ¼ of jaw.	- Incisors : canines : premolar : molars - Primitive mammals – 3:1:4:3 - Primitive primates – Dental comb and 36 teeth - Monkeys, apes, humans – 32 teeth 2:1:2:3 - General reduction in prognathism (jaw size)

		• Monkeys and apes have a gap called the diastema – canines • Canines are usually longer than other teeth • 3 cusped molars in early mammals • 5 cusped molars in humans and apes due to diet of fruit
	Dental Arcade 	• Apes – U shaped jaw • Humans – parabolic (V shaped) • Apes and australopithecines have larger teeth and distinct gap between canines and incisors. Homo habilis had smaller, narrower teeth, but canines still prominent. This reflects change in diet to softer foods, e.g. greater amount of meat and eventually cooked food in homo erectus. • From homo erectus to homo sapiens, teeth size has decreased, so the jaw is less prognathic, leading to parabolic dental arcade.
Nose	Olfaction	• Reduced sense of smell with gradual reduction in size of snout • There is a reduction in prognathism in humans compared to apes • Leads to a flatter face • Skills are becoming smaller and flatter – sense of smell becomes less important when other senses such as sight become more important. • In apes and early Hominins the jaw is more prognathic.
	Protrusion of Nose	• Rest of the face has flattened and there is less prognathism in humans, the nose has not flattened • In apes with pronounced prognathism – nose is flat • Leads to a flatter face

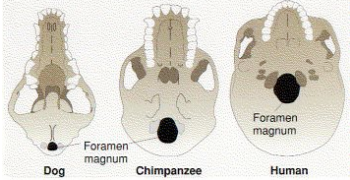
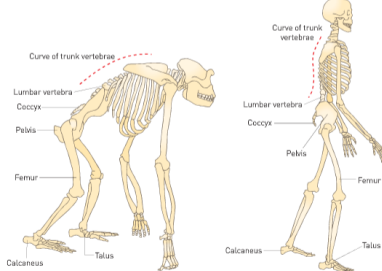
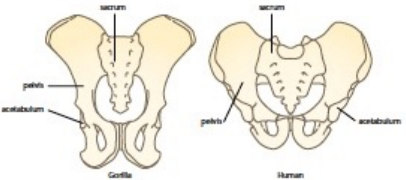
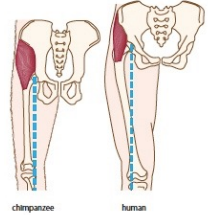
Hominin Evolutionary Trends

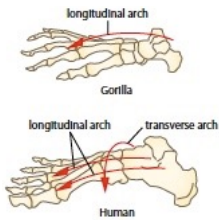
Hominin: classification is subfamily. It includes all ancestors of humans and *H. sapiens*.

Adaptations for erect posture

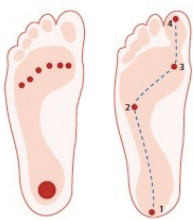
- Human ancestors were all bipedal
- Footprints from human ancestor found in Laetoli, Tanzania which were over 3 million years old. These footprints were from an early species (likely *a. afarensis*) which had adapted to upright, bipedal form of locomotion.



Structure	Adaptation
Foramen magnum	- Hole in the skull where brain joins onto spinal cord. - Quadrupeds (all 4s) – located towards the back of the skull, large muscles required to hold head in position. - Humans – located centrally underneath the skull - Allows skull to balance on top of vertebral column, which bears the weight of the skull 
Jaw bone	- Apes – protruding jaw, more weight in front of spine than behind - Humans – reduction in size and protrusion of jaw, giving flatter facial profile - Allows for balance of skull on top of the spine as the weight in front of and behind the foramen magnum is approx. equal.
Vertebral column	- Apes had a 'C-shaped' curve – most of the weight in front of the vertebral column - Humans 'S-shaped' curve, wedge-shaped lumbar vertebrae, forming forward-jutting curve - Improves balance in the upright position, allowing head to balance on top of neck. Cervical curve in neck brings vertebral column directly under the centre of gravity of the skull  Figure 18.3 Posture of the skeletons of a gorilla and a human
Pelvis	- Apes – narrow, longer from top to bottom - Humans – broader, shorter from top to bottom, bowl shaped - Shape supports abdominal organs in upright stances and attachment of large buttock muscles that move the legs and keep upper body erect 
Femurs (carrying angle)	- Apes – narrow pelvis, hip sockets close together, femur attached vertically to pelvis, weight transmitted on the inside of the femur - Humans – as the pelvis is broad, the hip sockets (acetabulum) are wide apart; the femurs tend to converge towards the knees, weight transmitted on outside of femur - Arrangement of femurs forms an angle known as carrying angle - Ensures the distribution of weight remains close to central axis of body and greater stability when walking 

Knee joint	- Humans – knee is a two-part hinge, the outer hinge is larger and stronger than the inner one as this is where the weight is transmitted - Strong ligaments prevent the knee bending backward, no energy required to support the joint in a standing position - Outer 'hinge' larger and stronger, to take weight of body. Knee is able to be straightened
Foot	- Apes – prehensile feet, opposable first digit, longitudinal arch only - Humans – no prehensility, big toe is large and aligned with other toes, two arches: longitudinal and transverse - Enlarged talus (heel bone) - Enable balance and stability in bipedal locomotion, allowing the striding gait – walking where knees and hips are fully extended 
Centre of gravity	- Apes – long arms, shorter legs. Centre of gravity is at chest level - Humans – short arms, longer legs. Centre of gravity is at pelvis level - Long legs increase length of stride and lower centre of gravity, contributing to stability when walking bipedally or standing upright

Stance and locomotion



- Humans walk where knee and hip joints are fully extended
- Muscle tone aids with this locomotion – partial contraction of skeletal muscle
 - E.g. to keep the head erect, the muscles in the back of the neck are partially contracted to stop it slumping forward
- Opposability of big toe has been lost as it is now weight bearing
- Forward swinging of the arms compensates for the natural rotation that occurs in pelvis

Advantages of bipedalism

- Seeing over tall grass to see predators or locate carcasses (scavengers) at a distance
- Increased size deters predators
- Carrying food while walking
- Efficient locomotion – efficient means of moving across the growing expanses of savannah
- Holding tools and weapons
- Thermoregulation – walking upright exposes 60% less surface area to the sun at midday and there is greater airflow across the body when it is lifted higher off the ground

Brain size and organisation:

Larger cranial capacity – 1350 cm³.

Skull shape:

Foramen magnum closer to centre of skull

Facial features:

Forehead, no brow ridge, protruding nose, less prognathism.

Teeth size and shape:

Decrease in size of teeth. Dental arcade parabolic rather than 'U' shaped.

Spine and pelvis shape (male and female):

Lumbar and cervical arch create 'S' shaped spine. Wider, shorter pelvis.

Hands (degree of prehension, dexterity):

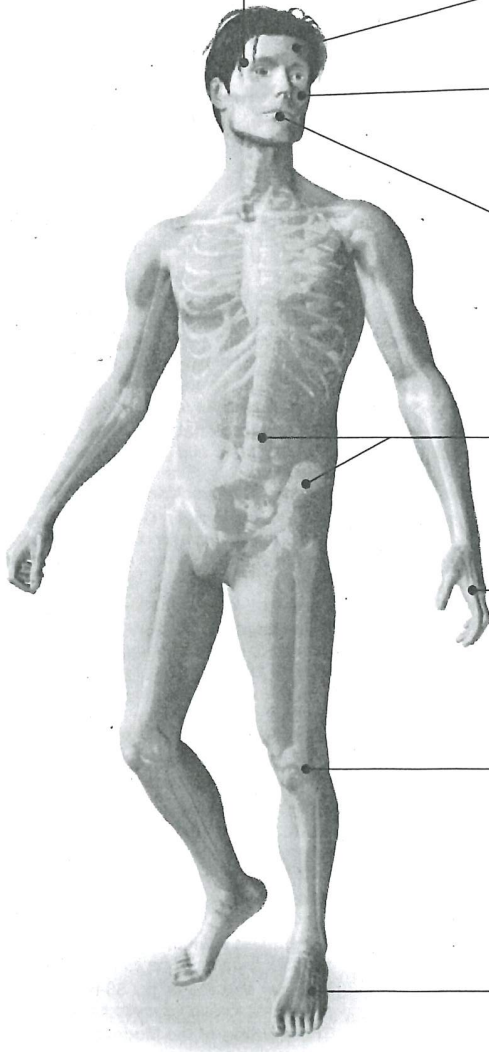
Opposable thumb, power and precision grip.

Leg shape, hip joint (carrying angle) and knee joint:

Acetabulum wider apart, femur carrying angle bears weight on outside of femur, distribution of weight remains central.

Feet adaptations to bipedalism:

Longitudinal and transverse arch. Non-opposable big toe, no prehensility, enlarged talus.



Evolutionary trends in hominins

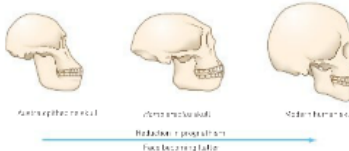
A.: stands for Australopithecus

P.: stands for Paranthropus

H.: stands for homo

Hominin	Age, year and site discovered	Cranial capacity and skull shape	Anatomical Features	Cultural development	Tools
A. afarensis	3.7-2.5 mya 1974 at Hadar in Ethiopia Laetoli footprints 	 Cranial capacity 430cc Foramen magnum at back, brow ridge and flat nose.	More human like than ape like. First hominids to walk bipedally, carrying angle of femur more human like, had an arched foot. Height approx. 1.5m Very prognathic jaw, intermediate shape between 'U' and 'V' shaped teeth. Teeth smaller than apes but maintain diastema.	- Trees as refuge and source of food - Lived in groups - Home bases - Sexual dimorphism - Herbivores - Environment – woodland, savannah	Oldowan tools Made from river-worn pebbles with flakes being removed. Flakes knocked from several angles to produce a core with cutting edge. 
A. africanus	3 mya 1924 at Limestone Quarry in Taung, South Africa	 Cranial capacity 457cc Foramen magnum at back, brow ridge and low facial angle.	Gracile form Australopithecine. Lightly built. Very prognathic jaw and human-like 'V' shaped teeth. Very large molars.	Herbivore	Oldowan tools 
P. robustus	2-1 mya 1938 at Kromdraai Cave in South Africa Died out because of change to climate	 Cranial capacity 542cc Little of skull behind foramen magnum. Very prominent sagittal crest. Heavy brow ridge, and large cheek bones.	Robust form, heavy set. Prognathic. Human-like teeth, canines smaller than apes, larger than humans. Midway between 'U' and 'V' shape. Heavy, robust teeth.	- Herbivore - Spent most time foraging for food	Oldowan tools 

H. habilis	2-1.7 mya 1960 at Olduvai Gorge in Tanzania	 Cranial capacity 590cc Bulge in area of brain for speech production. More of the skull is behind the foramen magnum.	Reduction in prognathism. Narrowed teeth crowns and more robust hands than humans – indicated still tree climbing.	- Meat incorporated into diet - Scavengers – males scavenged and females gathered - Shared food	Oldowan tools 
H. erectus	1.8-0.03 mya (30 000 years) 1891 at Island of Java in Indonesia	 Cranial capacity 1004cc Increase may be because of eating meat. Cranium thick boned. More of skull is behind the foramen magnum.	Reduction in prognathism, pronounced brow ridge. Angular protrusions on skull for muscular attachment for neck muscles. Relied on vision not olfaction. Intermediate dental arcade – more parabolic shape.	- Used fire - Hunting – men hunted and women gathered - Meat in diet - Lived in large areas - Home bases, seasonal campsites - Language - Social groups and division of labour	Acheulian hand axe 'Tear drop' shape. Ranged greatly in size – often referred to as hand axes. More flakes removed from the axe. 
H. neanderthalensis	0.2-0.03mya (200 000 – 30, 000 years) 1856 at Neander Valley in Germany	 Cranial capacity 1485cc Cranium elongated with slight forehead Brow ridge and large nose Central foramen magnum.	Short muscular structure	- VERY COLD conditions - Caves/shelters - Group cooperation to hunt LARGE animals - Rudimentary language - Social burial of dead, rituals, look after disabled, cannibalism	Mousterian flake tool Made of flint – predictable way of chipping. Preparation of core and striking of large oval flake 
H. sapien	0.2mya – present (200, 000 years) 1868 at Cro-Magnon in France	 Cranial capacity 1350cc Shorter, smoother, dome-shaped skull. Small, flat face and enlarged frontal cortex. Central foramen magnum.	Flatter face, lacks brow ridge. Jaws and teeth relatively narrow. Has chin.	- Improved hunting techniques - Technology of stone tools on mass (stone, bone, ivory, antler) - Ritualistic behaviour - Artistic expression (art, jewellery, music) 10, 000 years ago hunter/gatherer to agricultural society	Aurignacian – stone industries   Solutrean tools – spear heads  Magdalenian – fishing tools 

Trend		Increase in average brain size. Cerebral cortex surface area is increased by number of convolutions. Increase in surface area leads to higher functioning such as thinking, reasoning, planning and processing Overall increase in the size of the cranium.	Reduction in prognathism – face becoming flatter. Decrease in brow ridge and reduction in snout gives a flatter face  Decrease in size of incisors and canines Decrease in size of molars – progressive change in diet to softer foods Dental arcade has become more parabolic 		
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